Particle Physics

Based on lectures by Thomas Ryttov

Notes taken by Mikkel Kielstrup Holst

These notes are not endorsed by the lecturers. All errors are mine.

Contents

1. Introduction

As physicists we want to figure out what matter is made of. This is what particle physics attempts to do. We will find that the fundamental building blocks of matter can be classified into a relatively small number of types.

This course tries to get a qualitative understanding of *the Standard model*, which is our current best theory for understanding the nature of particles and their interactions. We will not do any explicit quantum field theory, but we will cover *Feynman diagrams* and their associated *Feynman rules*.

2. The Zoo

2.1. A Lesson in History

J. J. Thomson discovered the electron e^- in 1897 and gave birth to particle physics. Later in 1911 E. Rutherford discovered that atoms contain a massive positive nucleus. As should be familiar the nucleus of hydrogen is the proton p^+. The last of the three *classical particles* was found in 1932 when Chadwick discovered the neutron n.

M. Planck quantized the electromagnetic radiation in 1900 by assuming it came in small packages of energy $E = \hbar\omega$. A few years later in 1905 A. Einstein explained the photoelectric effect by assuming the electromagnetic field itself was quantized. This was radical at the time since the wave theory was extremely succesful. The matter was partly resolved in 1923 when A. H. Compton showed that light scattered from a particle at rest experiences a shift in wavelength by

$$\lambda' = \lambda + \frac{h}{mc}(1 - \cos\theta) \sim \text{Compton scattering}$$

This is exactly what you get if you treat light as a particle with $E = \hbar\omega$ and $m_\gamma = 0$. We now call this particle the photon γ.

The classical model has a problem, since it does not explain why the nucleus is held together. This new force is called the *strong force* and it must be short ranged! The first reasonable attempt at explaining the strong force came in 1934 by H. Yukawa when he suggested it was due to the exhange of a particle we now call the pion π. The π is an example of a *meson* whereas the e^- is a *lepton* and p^+, n are *baryons*. In 1947 the π is found in cosmic rays by C. Powell along with the muon μ which can be treated as a heavier e^-. As we will see there are actually three π: π^0 and $\pi^\pm$.

P. A. Dirac predicted the existence of *anti-particles* in 1930. With the positron e^+ being found the year after by C. D. Anderson. Following this many more anti-particles were found e.g. the anti-muon μ^+, anti-proton p^- and anit-neutron $\overline{n}$. Some particles are their own anti-particles such as the photon $\overline{\gamma} = \gamma$ and π^0.

Consider the process

$$A + B \to C + D$$

Due to *crossing symmetry* the following reactions are also allowed

$$A \to \overline{B} + C + D$$

$$A + \overline{C} \to \overline{B} + D$$

$$\overline{C} + \overline{D} \to \overline{A} + \overline{B}$$

However some might be forbidden kinematically!

As an example consider Compton scattering

$$\gamma + e^- \to \gamma + e^-$$

This is *in a way* the same process as pair annihilation

$$e^- + e^+ \to \gamma + \gamma$$

Though they appear completely different in experiment.

Consider β-decay as in

$$A \to B + e^-$$

Due to *charge conservation* we require $Q_A = Q_B - 1$ and we now know $A = n$ and $B = p^+$. This process has problems since the e^- is shown in experiment to be emitted with too little energy. This was resolved in 1930 by W. Pauli who suggested the existence of the neutrino ν and anti-neutrino $\overline{\nu}$ making β-decay

$$n \to p^+ + e^- + \nu/\overline{\nu}$$

Similarly people began suggesting that ν was involved in other decays such as that of π and μ in cosmic rays

$$\pi \to \mu + \nu/\overline{\nu}$$

$$\mu \to e^- + 2\nu/\overline{\nu}$$

Where we know there are two ν in the μ-decay since the energy of the e^- varies. The existence of ν and $\overline{\nu}$ was hard to prove as they are neutral and only interact weakly.

To determine whether ν or $\overline{\nu}$ are involved in a given process we define the *lepton number* by

$$L(e^-, \mu^-, \nu) = +1; \quad L(e^+, \mu^+, \overline{\nu}) = -1; \quad L(\text{other}) = 0$$

and propose the *law of conservation of lepton number*. Then we can write

$$\begin{aligned} n &\to p^+ + e^- + \overline{\nu} \\ \pi^- &\to \mu^- + \overline{\nu} \\ \pi^+ &\to \mu^+ + \nu \\ \mu^- &\to e^- + \nu + \overline{\nu} \\ \mu^+ &\to e^+ + \nu + \overline{\nu} \end{aligned}$$

This is what distinguishes ν from $\overline{\nu}$. However the process

$$\mu^- \to e^- + \gamma$$

never occurs even though no rules seem to broken. This implies the conservation of lepton number *within each generation* and the existence of multiple ν. We call these ν_e and ν_μ. The lepton number is assigned as before but now we have an *electron number* L_e and *muon number* L_μ. Then we have

$$\begin{aligned} n &\to p^+ + e^- + \overline{\nu}_e \\ \pi^+ &\to \mu^+ + \nu_\mu \\ \mu^- &\to e^- + \overline{\nu}_e + \nu_\mu \end{aligned}$$

etc. A similar law is the *law of conservation of baryon number*.

The first *kaon* K^0 was discovered in 1947 through

$$K^0 \to \pi^+ + \pi^-$$

and a couple years later in 1949 K^+ was found through

$$K^+ \to \pi^+ + \pi^+ + \pi^-$$

That K^+ was a charged version of K^0 was only realised much later. Since they behave like heavy π they are categorized as mesons. More mesons such as ρ^0 and ϕ were also discovered. At the same time Λ was found through

$$\Lambda \rightarrow p^+ + \pi^-$$

For the baryon number to be conserved Λ must be a baryon. More baryons such as Σ^0 and Δ^+ were also discovered. All these particles also came with their own anti-particles.

These new particles are called *strange* since they are produced in large amounts at a scale $\sim 10^{-23}$ s but decay slowly at a scale $\sim 10^{-10}$ s. A. Pais suggested that this was due to their production and disintegration mechanisms being different. We say they are produced by strong interactions and decay by weak interactions. He also suggested that strange particles were produced in pairs. This idea lead to M. Gell-Mann and K. Nishijima giving each particle a *strangeness*. This new property is conserved in strong and electromagnetic interactions but violated in weak interactions. We assign the following

$$S(p,n) = 0; \quad S(\Sigma s, \Lambda) = -1; \quad S(\Xi^-, \Xi^0) = -2$$

$$S(\eta, \pi s) = 0; \quad S\left(K^0, K^+\right) = 1; \quad S\left(\overline{K}^0, K^-\right) = -1$$

The lines above and all other particles can be arranged into geometric shapes according to *the eightfold way*. This is in many ways the periodic table of particle physics, at least for the *hadrons*.

To explain the patterns of the eightfold way Gell-Mann and G. Zweig proposed that hadrons consisted of *quarks* q. These come in three *flavors*: up u with charge $\frac{2}{3}$ and $S = 0$, down d with charge $-\frac{1}{3}$ and $S = 0$, and strange s with charge $-\frac{1}{3}$ and $S = -1$. Each has an anti-quark $\overline{q}$ with opposite charge and strangeness. All baryons are then composed of three quarks while all mesons are composed of a quark and an anti-quark. With the above we can construct ten baryons and nine mesons. These constitute the *baryon decuplet* and the *meson nonet*. To construct more supermultiplets we need to include spin which we will see later.

There are two problems with the quark-model. The first problem is that we have never observed a quark. The only experimental evidence we have is that the proton has substructure indicating the existence of quarks. The second is that they appear to violate the *Pauli exclusion principle* since e.g. Δ^{++} contains three u with spin $\frac{1}{2}$. To solve this O. W. Greenberg proposed that quarks carry an additional degree of freedom. We call this *color* and quarks can be red, green and blue. All particles we see in nature must be *colorless* meaning they have zero color or equal amounts of each.

2.2. The Standard Model

As seen in the previous section a bunch of particles were found. The basic building blocks being *leptons* and *quarks*

$$\text{quarks:} \quad [u, d, c, s, t, b]$$

$$\text{leptons:} \quad [e, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau]$$

With those not mentioned previously eventually being found.

The weak interaction is mediated by the *intermediate vector bosons* W^+/W^- and Z, the electromagnetic interaction is mediated by photons γ, and strong interaction is mediated by eight *gluons*.

The final particle found in 2012 was the *Higgs boson* which is responsible for particles having mass!

This is the *Standard Model*.

3. Dynamics Overview

As far as we know nature is governed by four fundamental forces: *strong*, *electromagnetic*, *weak*, and *gravitational*. We will only consider the first three as quantum gravity is a different beast which has yet to be conquered.

3.1. QED

Quantum electrodynamics is the theory that describes the electromagnetic force and is by far the most simple and best understood of the aforementioned theories.

All electromagnetic phenomena are reducible to the process in Figure 1

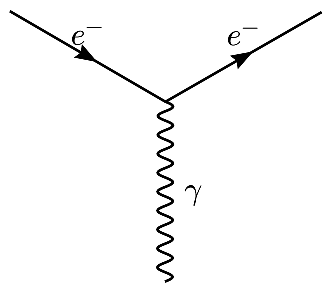


Figure 1: A charged particle e^- *enters and emits or absorbs a photon* γ *before exiting. This is an example of a* Feynman diagram.

This is the only allowed vertex in QED and by patching these processes together we get more complicated interactions. An example is *Møller scattering* seen in Figure 2

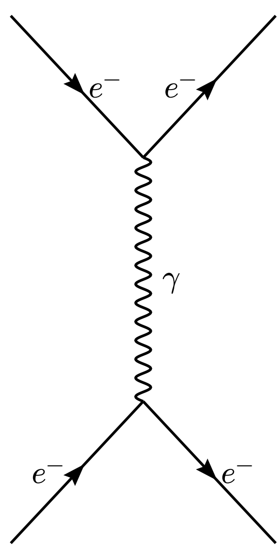


Figure 2: Feynman diagram for Møller scattering. Two electrons e^- *scatter by exchanging a photon* γ.

Another example is *Bhabha scattering* seen in Figure 3

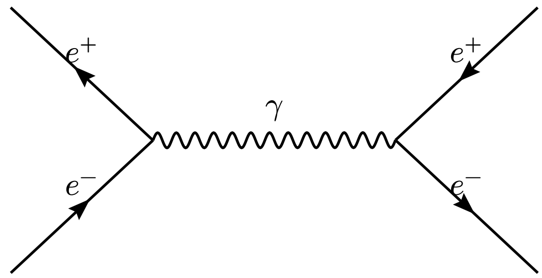


Figure 3: Feynman diagram for Bhabha scattering. An electron e^- and positron e^+ annihilate to form a photon γ which produces a new electron-positron pair.

This is just the diagram for Møller scattering on its side! We interpret the particles going *backwards* in time as antiparticles. A second diagram also contributes to Bhabha scattering. This diagram is seen in Figure 4

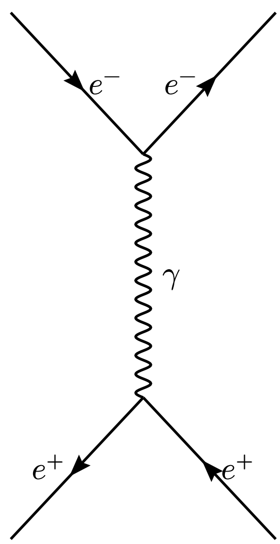


Figure 4: Feynman diagram for Bhabha scattering. An electron e^- and positron e^+ scatter by exchanging a photon γ.

As a final example we consider *Compton scattering* seen in Figure 5

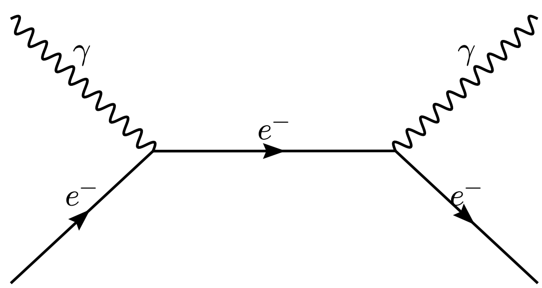


Figure 5: Feynman diagram for Compton scattering.

Almost identical diagrams can be made for *pair annihilation* and *pair production.*

We could allow additional vertices in the above to get more complicated diagrams. However each additional vertex comes with a factor of α being the *fine-structure constant* meaning diagrams with additional vertices are less important. Since the observed process is only represented by *external lines.* The *internal lines* represent *virtual particles* and describes the responsible mechanism. The Feynman diagram is a *purely symbolic tool*! We compute things from Feynman diagrams using *Feynman rules* which we will discuss later. These rules enforce energy and momentum conservation at each vertex.

3.2. QCD

Quantum chromodynamics is the theory that describes the strong force. In QCD *color* plays the role of charge and the fundamental process is seen in Figure 6

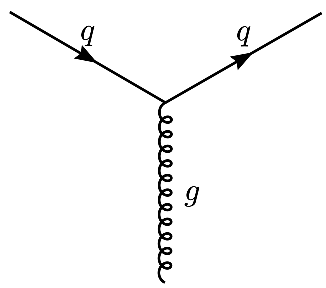


Figure 6: Primitive vertex *for QCD.*

We can again patch these together to form more complication interactions. An example is seen in Figure 7

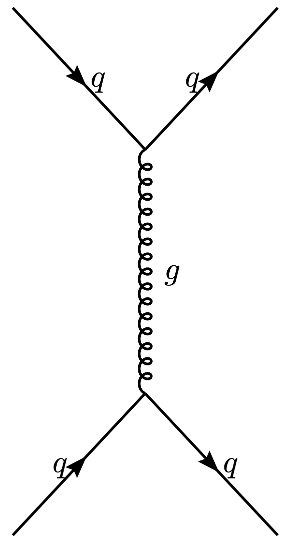


Figure 7: Feynman diagram for quark binding.

There exists three types of color and quarks can change color with the difference being carried by gluons. An example is seen in Figure 8

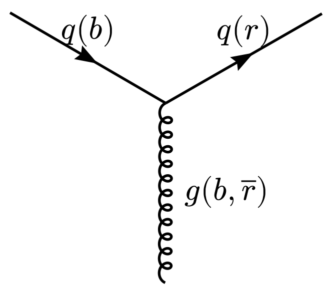


Figure 8: Conversion of a blue quark into a red quark with the help of a bicolored *gluon. This implies the existence of nine different gluons. However there are only eight.*

Due to gluons being colored they also couple to themselves. (unlike the photon γ which is neutral!) We get two new kinds of vertices as seen in Figure 9

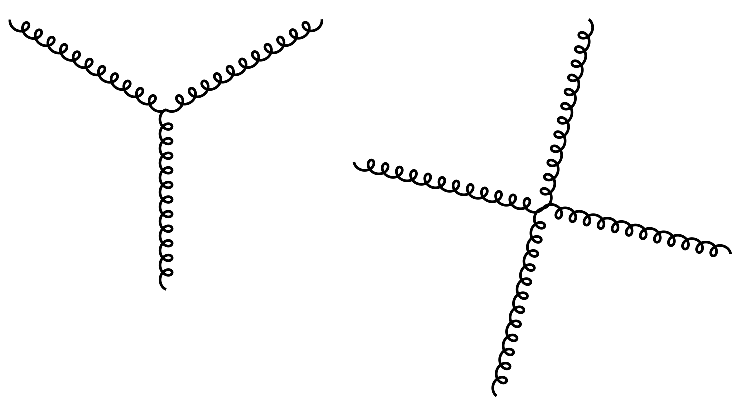


Figure 9: Two additional kinds of primitive vertices in QCD.

This additional coupling necessarily makes QCD way more complicated than QED.

3.3. Weak interactions

All quarks and leptons carry a *weak charge* meaning they can all interact through weak interactions. We distinguish between *charged* interactions mediated by $W^{\pm}$ and *neutral* interactions mediated by Z.

The fundamental neutral vertex is seen in Figure 10

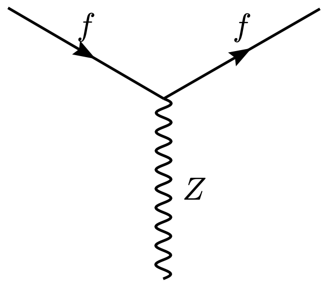


Figure 10: The fundamental neutral vertex. Here $f \in \{q, l\}$

Then for any process mediated by γ there is an analogous process mediated by Z. An example of a more complicated interaction is seen in Figure 11

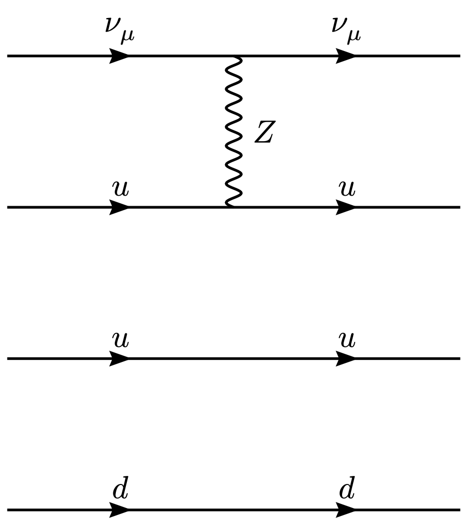


Figure 11: The scattering of a muon-neutrino ν_μ with a proton p. Here the two up-quarks u are called spectator quarks.

The fundamental charged vertices are seen in Figure 12

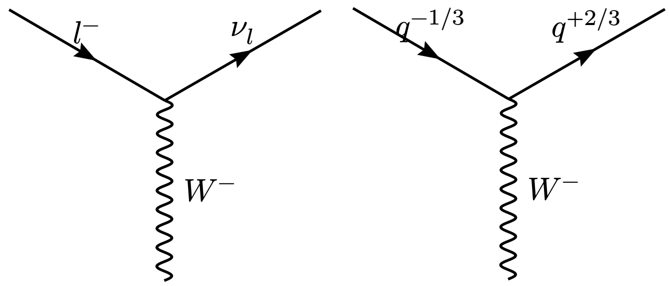


Figure 12: The fundamental charged vertices. The opposite direction is mediated by W^+. The quarks in the second diagram have the same color but different flavors since flavor is simply not conserved.

An example of a very important process is seen in Figure 13

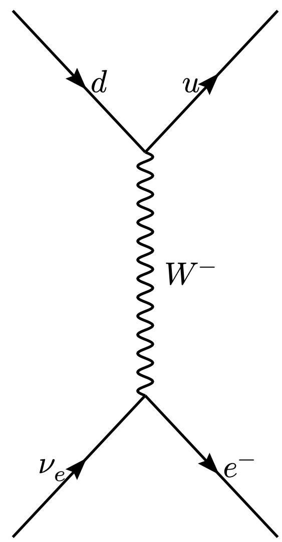


Figure 13: The semileptonic *process $d + \nu_e \to u + e^-$.*

This diagram gives us β-decay $n \to p + e^- + \overline{\nu}_e$! Since $udd \to uud$ with two spectator quarks.

The above picture is a bit simplistic since it seems like the weak interaction only couple pairs $\{u, d\}$. What actually happens is the coupling of pairs $\{u, d'\}$ with

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

This is called the *CKM-mechanism* and accounts for the decay of Δ and Ω^-.

The $W^\pm$ and Z also couple directly and since $W^\pm$ is charged it also couples to photons γ. These are all seen in Figure 14

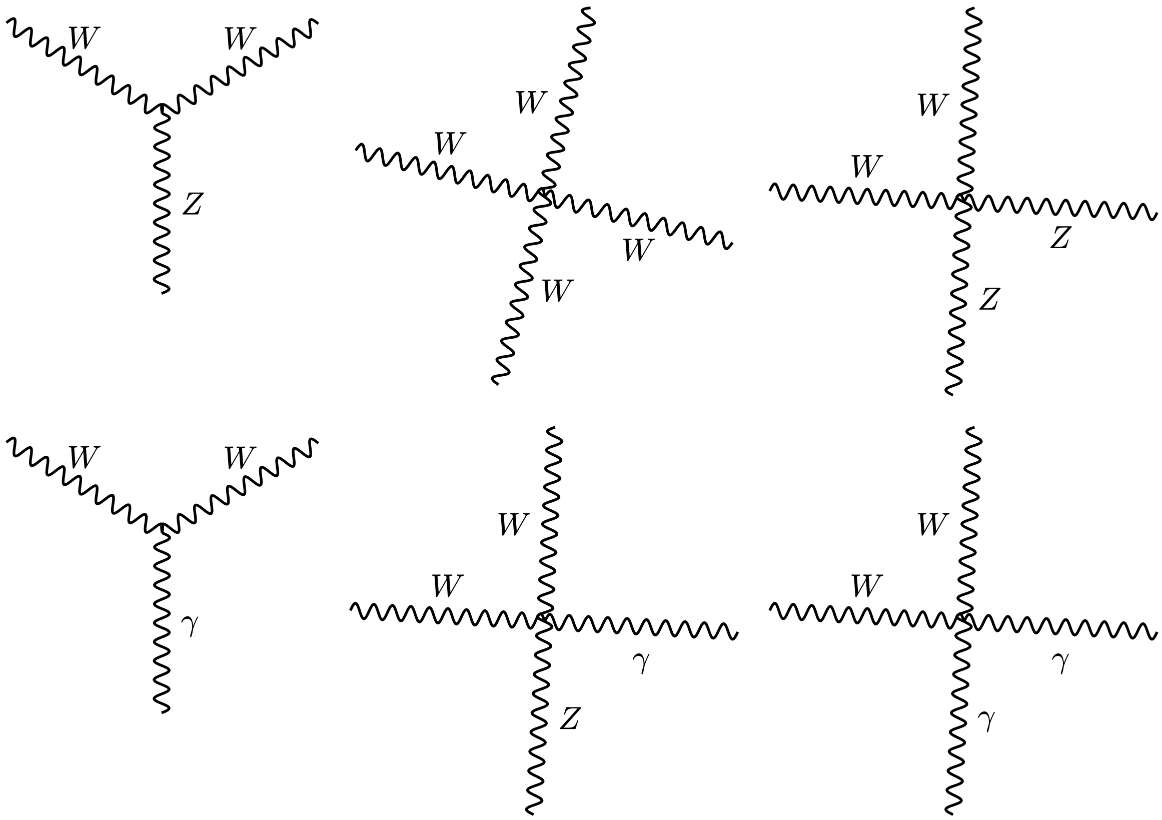


Figure 14: The couplings between $W^{\pm}, Z$ and γ.

These are not super important for us however.

3.4. Decays and laws

All particles want to disintegrate into some lighter particles unless some law prevents it. The γ is stable since $m_\gamma = 0$ while e^- is stable since it is the lightest charged particle so by charge conservation it can not decay. Similarly the p is stable due to baryon number conservation. However most particles want to decay and most *exotic* particles only exist in relatively small amounts.

Any decay is governed by one of the three interactions described above. Some examples are

$$\Delta^{++} \to p + \pi^+ \quad \text{strong interaction}$$

$$\pi^0 \to \gamma + \gamma \quad \text{electromagnetic interaction}$$

$$\Sigma^- \to n + e^- + \overline{\nu}_e \quad \text{weak interaction}$$

Some interactions are less obvious

$$\Sigma^- \to n + \pi^- \quad \text{weak interaction}$$

$$\Delta^- \to n + \pi^- \quad \text{strong interaction}$$

We have seen all fundamental vertices and by seeing what is conserved in these we can figure out what is conserved in general. We have conservation of

1. charge.
2. color.
3. baryon number (number of quarks).
4. lepton number.
5. flavor in strong and electromagnetic.

3.5. The Higgs particle

We briefly mention the fundamental vertices of the Higgs particle for completeness. The Higgs particle is neutral and can couple with itself as in Figure 15

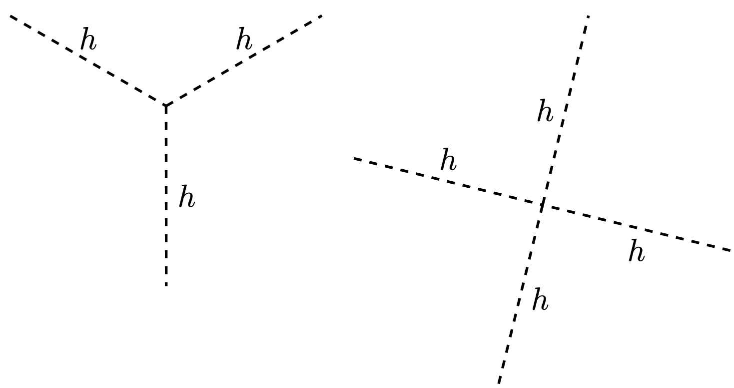


Figure 15: Self-coupling of the Higgs particle.

It also couples to $W^{\pm}$ and Z as in Figure 16

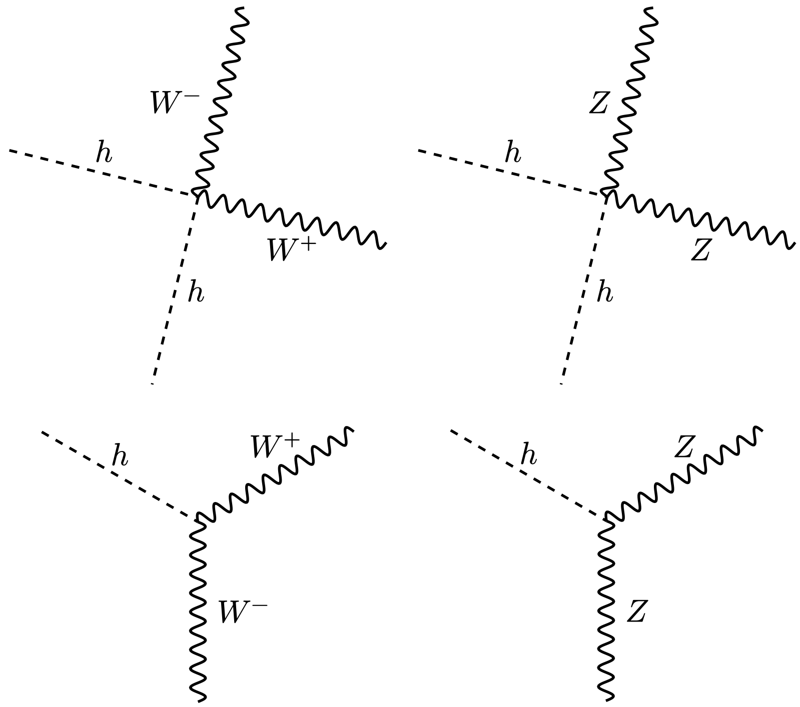


Figure 16: The couplings between $W^{\pm}, Z$ and h.

We also have an interaction with fermions as in Figure 17

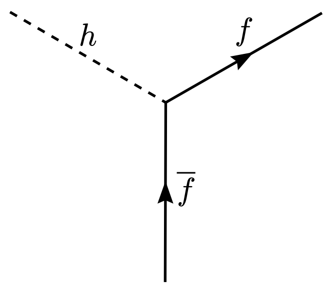


Figure 17: Coupling between fermions and the Higgs particle.

We care about the Higgs particle since the interaction in Figure 17 is dependent on m_f implying the mass of a given particle is dependent on how much they interact with the Higgs particle.

4. Special Relativity

4.1. The Lorentz transformations

According to *special relativity* then the laws of physics are valid in all *inertial* frames.[1]

Consider two inertial frames S and S', with S' having a uniform velocity $\boldsymbol{v}$ relative to S. We use coordinates where $\boldsymbol{v}$ is along $\hat{x}$. The frames are then related by

$$x' = \gamma(x - \beta ct); \quad y' = y; \quad z' = z; \quad ct' = \gamma(ct - \beta x)$$

with

$$\gamma \equiv \frac{1}{\sqrt{1-\beta^2}} \sim \text{Lorentz factor}$$

$$\beta \equiv \frac{v}{c}$$

These are the *Lorentz transformations*. The inverse transformations are given by $\beta \to -\beta$. These transformations have many consequences such as the *relativity of simultaneity*, *Lorentz contraction*, and *time dilation*. We care about *velocity addition*

$$u = \frac{u' + v}{1 + u'v/c^2}$$

where u and u' is the velocity of some object in S and S' respectively.

4.2. Four-vectors

We define the *position four-vector* x^μ

$$x^\mu = (ct, x, y, z)$$

By comparison with above the Lorentz transformations become

$$x'^0 = \gamma\left(x^0 - \beta x^1\right), \ \text{etc.}$$

We write $x'^\mu = \Lambda^\mu{}_\nu x^\nu$[2] where $\Lambda^\mu{}_\nu$ represents the Lorentz transformations for a given $\boldsymbol{v}$. We define a general four-vector A^μ as an object transforming like x^μ

$$A^\mu = \Lambda^\mu{}_\nu A^\nu$$

[1] By inertial we mean non-accelerating.
[2] Using the *summation convention*.

We would like *invariant* quantities. These can be found by *contractions*[3] using the *metric* $g_{\mu\nu}$

$$\text{scalar} = g_{\mu\nu} A^\mu A^\nu \equiv A_\mu A^\mu \equiv A^2$$

We work in flat *Minkowski* spacetime where $g_{\mu\nu} = \eta_{\mu\nu}$.[4] The *spacetime interval* is the most important example

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu = c^2\,\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2$$

We can extend the definition of a four-vector to define general *second-rank tensors* $T^{\mu\nu}$ as objects transforming like

$$T'^{\mu\nu} = \Lambda^\mu{}_\kappa \Lambda^\nu{}_\sigma T^{\kappa\sigma}$$

The definition of higher rank tensors is obvious. We can raise and lower indices as implied before using $g_{\mu\nu}$

$$T^\mu{}_\nu = g_{\nu\kappa} T^{\mu\kappa} \sim \text{mixed}$$

We can also use $g_{\mu\nu}$ to take the trace

$$T^\mu{}_\mu = g_{\mu\nu} T^{\mu\nu} \sim \text{scalar}$$

4.3. The four-momentum

We define the *proper time* τ by

$$\mathrm{d}\tau = \frac{\mathrm{d}t}{\gamma}$$

This is the time measured in the *instantaneous rest-frame*.[5]

We define velocity as

$$\boldsymbol{v} \equiv \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} \sim \text{lab frame}$$

Then the *proper velocity* is defined by

$$\boldsymbol{\eta} \equiv \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\tau} = \gamma \boldsymbol{v}$$

[3] All indices *cancel*.

[4] We use $\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$.

[5] We use $\mathrm{d}s^2 = c^2\,\mathrm{d}\tau^2 = c^2\,\mathrm{d}t^2 - \mathrm{d}\boldsymbol{r}^2$.

This quantity is useful since $\mathrm{d}\tau$ is a scalar. This motivates the *four-velocity* η^μ

$$\eta^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$$

The related invariant is $\eta_\mu \eta^\mu = c^2$. We can now define the *four-momentum* p^μ

$$p^\mu \equiv m\eta^\mu$$

With $E \equiv \gamma mc^2$ we can write p^μ as

$$p^\mu = \left(\frac{E}{c}, \boldsymbol{p}\right)$$

with $\boldsymbol{p} = \gamma m\boldsymbol{v}$. The related invariant is

$$p_\mu p^\mu = \frac{E^2}{c^2} - \boldsymbol{p}^2 \overset{\text{rest frame}}{=} m^2c^2$$

We can expand E to obtain

$$E = \underbrace{mc^2}_{\text{rest energy}} + \underbrace{\frac{1}{2}mv^2 + \dots}_{\text{kinetic energy}}$$

When $m = 0$ the above definitions of E and $\boldsymbol{p}$ break and become *indeterminate* when $v = c$. This implies particles with $m = 0$ can exist provided their $v = c$ always. The energy is then determined from $E = \hbar\omega$ with $\boldsymbol{p}$ given by the invariant above.

We care about p^μ since it is *always* conserved in any physical process!

4.4. Examples

Consider two lumps with mass m colliding at $\frac{3}{5}c$. After the collision the lumps stick together.

We obviously have $\boldsymbol{p}_1 = -\boldsymbol{p}_2$ meaning $\boldsymbol{p}_M = 0$. Then since $E_1 + E_2 = E_M$ we have

$$Mc^2 = 2E_1 = \frac{2mc^2}{\sqrt{1 - \left(\frac{3}{5}\right)^2}} = \frac{5}{2}mc^2$$

implying

$$M = \frac{5}{2}m > 2m$$

A particle of mass M at rest decays into two pieces with mass m.

By $\boldsymbol{p}_M = \boldsymbol{p}_1 + \boldsymbol{p}_2 = 0$ we have $\boldsymbol{p}_1 = -\boldsymbol{p}_2$. Then since $E_M = 2E_1$ we have

$$M = \frac{2m}{\sqrt{1-\beta^2}} \Rightarrow \beta^2 = 1 - \left(\frac{2m}{M}\right)^2$$

We need $M \geq 2m$ for the process to make sense. We refer to $M = 2m$ as the *threshold*.[6]

A π at rest decays into $\mu + \nu$.

We have

$$p_\pi = p_\mu + p_\nu \Rightarrow p_\nu = p_\pi - p_\mu$$

implying

$$p_\nu^2 = p_\pi^2 + p_\mu^2 - 2p_\pi \cdot p_\mu$$

Where $p_\nu^2 \sim 0$ and

$$p_\pi^2 = m_\pi^2 c^2; \quad p_\mu^2 = m_\mu^2 c^2; \quad p_\pi \cdot p_\mu = \frac{E_\pi E_\mu}{c^2}$$

Then

$$0 = m_\pi^2 c^2 + m_\mu^2 c^2 - 2m_\pi E_\mu \Rightarrow E_\mu \simeq \frac{m_\pi^2 + m_\mu^2}{2m_\pi} c^2$$

Similarly with $p_\mu = p_\pi - p_\nu$ we find

$$m_\mu^2 c^2 = m_\pi^2 c^2 - 2m_\pi E_\nu$$

using $E_\nu \simeq |\boldsymbol{p}_\mu| c$ we have

$$|\boldsymbol{p}_\mu| \simeq \frac{m_\pi^2 - m_\mu^2}{2m_\pi} c$$

A p collides with a p at rest by $p + p \to p + p + p + \overline{p}$. What is the threshold energy?

We define the *center-of-momentum* frame to be the frame where $\boldsymbol{p}_{\text{total}} = 0$. We consider p^μ_{total} before the collision in the *lab-frame*

$$p^\mu_{\text{total}} = \left(\frac{E + mc^2}{c}, |\boldsymbol{p}|, 0, 0\right)$$

and the p'^μ_{total} after the collision in the *CM-frame*

[6] When $\beta = 0$.

$$p'^{\mu}_{\text{total}} = \left(4m\frac{c^2}{c}, \mathbf{0}\right) \sim 4p \text{ at rest}$$

Then we use $p_\mu p^\mu = p'_\mu p'^\mu$ to obtain

$$E = 7mc^2$$

Consider some complicated S with

$$E_{\text{total}} = \sum_i \gamma_i m_i c^2; \quad \boldsymbol{p}_{\text{total}} = \sum_i \gamma_i m_i \boldsymbol{v}_i$$

Then

$$|\boldsymbol{p}_{\text{total}}| = \gamma\left(|\boldsymbol{p}_{\text{total}}| - \beta\frac{E_{\text{total}}}{c}\right)$$

The CM-frame S' is then given by

$$\beta = \frac{|p_{\text{total}}|c}{E_{\text{total}}}$$

We always have $\beta < 1$ meaning this frame always exists.

5. Symmetries

We care about *symmetries* in physics because of *Noether's theorem* which tells us that all continuous symmetries of an action S imply conserved currents.

5.1. Group theory

The notion of a symmetry is best described in the context of groups. A group is a set G along with a binary operation on G that combines any two elements $a, b \in G$ to form an element $a \cdot b \in G$ such that the *group axioms* are satisfied:

1. For all $a, b, c \in G$ we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
2. There exists an identity element $e \in G$ such that for all $a \in G$ we have $e \cdot a = a \cdot e = a$.
3. For each $a \in G$ there exists an element $b \in G$ such that $a \cdot b = b \cdot a = e$. We call $b = a^{-1}$ the inverse of a.

We call commuting groups *Abelian*. We can have *finite* or *infinite* groups, in particular we have *continuous* groups.[7]

The groups we care about in physics can typically be written as groups of matrices. We especially care about $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{O}(n)$, and $\mathrm{SO}(n)$.[8] Any group G can be *represented* by a group of matrices. This means each group element $a \in G$ has a corresponding matrix M_a with the correspondence respecting group multiplication,

$$M_a M_b = M_c$$

This representation is not necessarily *faithful* since multiple group elements can be represented by the same matrix.[9] We would say G_M is *homomorphic* to G. Any G which is already a group of matrices is obviously a representation of itself. We call this the *funda-*

[7]Within physics all such continuous groups are *Lie* groups.

[8]To be concrete $\mathrm{SO}(3)$ would describe rotational spatial symmetry and it appears almost identical to $\mathrm{SU}(2)$.

[9]As an example take $M_a = \mathbb{1}$ for all $a \in G$.

mental representation. But, we can also have representations of other dimensionalities. We can construct new representations by combining old ones

$$M_a = \begin{pmatrix} M_a^{(1)} & 0 \\ 0 & M_a^{(2)} \end{pmatrix}$$

However, these are boring and we only care about *irreducible* representations which cannot be decomposed as above.

5.2. Angular momentum

Within quantum mechanics we care about the orbital angular momentum $\boldsymbol{L}$ and the *spin* angular momentum $\boldsymbol{S}$. These angular momenta $\boldsymbol{J}$ are very similar and formally they are essentially the same. However, there are some differences. When performing measurements of $\boldsymbol{L}$ we find these are quantized[10]

$$L^2 \sim l(l+1)\hbar^2$$

$$L_z \sim m_l \hbar$$

where $l = 0, 1, ...$ and

$$m_l = -l, -l+1, ..., l-1, l$$

When performing measurements of $\boldsymbol{S}$ we similarly have

$$S^2 \sim s(s+1)\hbar^2$$

$$S_z \sim m_s \hbar$$

where $s = 0, \frac{1}{2}, 1, ...$ and

$$m_s = -s, -s+1, ..., s-1, s$$

All particles can be given any l, but each type of particle have their own spin s. We call particles with half-integer spin *fermions* and those with integer spin *bosons*.[11]

We can represent angular momentum states by $|lm_l\rangle$ and $|sm_s\rangle$. We would like to understand composite systems and how to handle the addition of angular momenta

[10] See e.g. *Sakurai* for details.

[11] All baryons, leptons, and quarks are fermions, while all mesons and mediators are bosons.

$$\boldsymbol{J} = \boldsymbol{J}_1 + \boldsymbol{J}_2$$

This could be the *total* angular momentum of a single particle

$$\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$$

or the spin of a composite system

$$\boldsymbol{J} = \boldsymbol{S}_1 + \boldsymbol{S}_2$$

This is made difficult since

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$$

implying we do not have access to all components at once. We are forced to work with a single component J_z and the magnitude J^2. We would like

$$|jm\rangle \sim |j_1m_1\rangle + |j_2m_2\rangle$$

The J_z add trivially so

$$m = m_1 + m_2$$

However, the J^2 is more annoying and we get all

$$j = |j_1 - j_2|, |j_1 - j_2| + 1, ..., j_1 + j_2 - 1, j_1 + j_2$$

We can explicitly decompose $|j_1m_1\rangle|j_2m_2\rangle$ as

$$|j_1m_1\rangle|j_2m_2\rangle = \sum_{j=|j_1-j_2|}^{j_1+j_2} \underbrace{C^{jj_1j_2}_{mm_1m_2}}_{\substack{\text{Clebsch-Gordan}\\ \text{coefficients}}} |jm\rangle$$

with $m = m_1 + m_2$.

5.3. Discrete symmetries

Consider the *parity* operator π which acts on a state $|\boldsymbol{x}\rangle$ as

$$\pi|\boldsymbol{x}\rangle = |-\boldsymbol{x}\rangle$$

Classically people believed physics were invariant under parity. However, weak interactions were shown to break parity in 1956 by Wu.

Consider a particle moving with velocity v we define the z-axis to be the direction of motion. We define the *helicity* as the value of

$$h \sim \frac{m_s}{s} = \pm 1$$

along this axis. When $h = +1$ the spin s and velocity v are parallel and we call the particle *right-handed*.[12] Otherwise, we call it *left-handed*. However, the value of h is not invariant since an observer moving with velocity greater than v would measure the opposite. The exception is particles moving with velocity c since their motion cannot be reversed. We can treat neutrinos as massless and by experiment we find[13]

$$\nu_L \sim \text{left-handed}$$

$$\overline{\nu}_R \sim \text{right-handed}$$

This occurs since weak interactions break parity. We can similarly determine the *handedness* of photons and we find an even split since electromagnetic interactions respect parity.

We characterize objects depending on how they transform under parity:[14]

$$\text{scalar} \sim \pi(s) = s$$

$$\text{pseudoscalar} \sim \pi(p) = -p$$

$$\text{vector} \sim \pi(\boldsymbol{v}) = -\boldsymbol{v}$$

$$\text{pseudovector} \sim \pi(\boldsymbol{a}) = \boldsymbol{a}$$

Also, by definition we have

$$\pi^2 = \mathbb{1}$$

implying the eigenvalues of π are ± 1. We associate

$$\pi(q) = +1; \quad \pi(\overline{q}) = -1$$

implying

$$\pi(B) = +1; \quad \pi\left(\overline{B}\right) = -1$$

To describe photons we use the vector potential A^μ so

[12] Also, note that h is odd under parity.

[13] This shows parity is *maximally* violated.

[14] We see a theory with only pseudovectors and scalars is invariant under parity. However, the addition of a vector breaks this invariance. This is exactly what happens in weak interactions.

$$\pi(\gamma) = -1$$

Consider the *charge conjugation* operator C which acts on a state $|p\rangle$ as

$$C|p\rangle = |\overline{p}\rangle$$

The operator C changes the sign of *all* internal quantum numbers.[15] Again by definition

$$C^2 = \mathbb{1}$$

implying the eigenvalues of C are ± 1. Then an eigenstate of C satisfies

$$C|p\rangle = \pm|p\rangle$$

which is only true for particles who are their own antiparticles. One can show a system consisting of a spin $\frac{1}{2}$ particle and its antiparticle in a configuration with orbital angular momentum l and total spin s, is an eigenstate of C with the eigenvalue $(-1)^{l+s}$.[16] We again have weak interactions violating invariance under C since

$$\nu_L \xrightarrow{C} \overline{\nu}_L \sim \text{does not exist}$$

We have seen that π is not respected due to

$$\pi^+ \to \mu^+ + \nu_{\mu,L}$$

and C is not respected since acting with C gives

$$\pi^- \to \mu^- + \overline{\nu}_{\mu,L}$$

which is impossible. However, we can act with both π and C. Then we obtain

$$\pi^- \to \mu^- + \overline{\nu}_{\mu,R}$$

which does occur! Thus we have found that physics appears to be invariant under CP. This has some weird consequences. We consider the process

$$K^0 \leftrightarrow \overline{K}^0$$

which is possible. This means the particles we typically observe are some linear combinations of K^0 and $\overline{K}^0$. The K's are pseudoscalars so

$$\pi\big|K^0\big\rangle = -\big|K^0\big\rangle; \quad \pi\big|\overline{K}^0\big\rangle = -\big|\overline{K}^0\big\rangle$$

[15]This leaves mass, energy, momentum, and spin unchanged.
[16]Think $M = q\overline{q}$.

Also,

$$C|K^0\rangle = |\overline{K}^0\rangle; \quad C|\overline{K}^0\rangle = |K^0\rangle$$

implying

$$CP|K^0\rangle = -|\overline{K}^0\rangle; \quad CP|\overline{K}^0\rangle = -|K^0\rangle$$

The normalized eigenstates of CP are then

$$|K_1\rangle = \frac{1}{\sqrt{2}}\big(|K^0\rangle - |\overline{K}^0\rangle\big); \quad |K_2\rangle = \frac{1}{\sqrt{2}}\big(|K^0\rangle + |\overline{K}^0\rangle\big)$$

with

$$CP|K_1\rangle = |K_1\rangle; \quad CP|K_2\rangle = -|K_2\rangle$$

Assuming CP is respected in weak interactions then K_1 can only decay into a state with $CP = +1$, while K_2 can only decay into a state with $CP = -1$. This implies[17]

$$K_1 \to 2\pi; \quad K_2 \to 3\pi$$

with the K_1 decay being much faster. Then we imagine a beam of K^0's

$$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_1\rangle + |K_2\rangle)$$

The K_1 component will decay quickly meaning at the end of our beam we should have a beam of pure K_2's. Then we expect many 2π decays near the source and many 3π near the end of our beam. The existence of K_1 and K_2 was shown experimentally and their lifetimes were determined to be

$$\tau_1 \sim 10^{-10} \text{ sec}$$

$$\tau_2 \sim 10^{-8} \text{ sec}$$

This allows us to test if CP is actually respected since given we have a long enough beam we can produce an arbitrarily pure sample of K_2's. Then if we still observed 2π decays CP would be violated. This means the long lived species is not an eigenstate of CP as we assumed, but instead contains a small amount of K_1

[17] Since each π carries $\pi = -1$.

$$|K_L\rangle = \frac{1}{\sqrt{1+|\varepsilon|^2}}(|K_2\rangle + \varepsilon|K_1\rangle)$$

with ε measureing the amount of CP violation. By experiment we determine

$$\varepsilon \sim 10^{-3}$$

Consider the *time reversal* operator T taking $t \to -t$. We care about this operator because of the CPT *theorem*: any Lorentz invariant local quantum field theory with an Hermitian Hamiltonian $H^\dagger = H$ respects CPT.[18] This implies T is violated since the combination CPT is respected. However, this is experimentally very difficult to test.

[18]The standard model is such a theory.

6. Computing Γ and σ

Within particle physics we can probe interactions through processes like decays and scattering. We would therefore like to be able to compute various physical quantities such as the lifetime of a particle.

6.1. Γ and σ

We define the *decay rate* Γ by[19]

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\Gamma N$$

implying

$$N = N_0 e^{-\Gamma t}$$

The mean lifetime τ is then

$$\tau = \frac{1}{\Gamma}$$

We typically have multiple decay modes for a given particle. With multiple decays we define the total decay rate Γ_{tot} by

$$\Gamma_{\text{tot}} = \sum_i \Gamma_i$$

and the lifetime is

$$\tau = \frac{1}{\Gamma_{\text{tot}}}$$

We can also define the *branching ratio* by

$$\text{branching ratio for } i^{\text{th}} \text{ mode} = \frac{\Gamma_i}{\Gamma_{\text{tot}}}$$

Which is a measure of how many particles decay through each mode. Then to compute both τ and the branching ratios we need to calculate Γ_i.

When discussing scattering we are interested in the *cross section* σ. All processes have their own cross section σ_i and we define the total cross section σ_{tot} by

[19] This is simply the probability per unit time that a particle will decay.

$$\sigma_{\text{tot}} = \sum_i \sigma_i$$

We now consider a particle encountering some potential and scattering off at an angle θ. This *scattering angle* will be a function of the *impact parameter* b. Classical *hard-sphere scattering* gives

$$b \overset{\text{hard-sphere}}{=} R\cos\frac{\theta}{2}$$

and *Rutherford scattering* gives

$$b \overset{\text{Rutherford}}{=} \frac{q_1 q_2}{2E}\cot\frac{\theta}{2}$$

We imagine a particle coming in with an impact parameter between b and $b + \mathrm{d}b$. This particle with then scatter with a scattering angle between θ and $\theta + \mathrm{d}\theta$. We can more generally consider a particle passing through an infinitesimal area $\mathrm{d}\sigma$ and scattering into a corresponding solid angle $\mathrm{d}\Omega$. They are related as

$$\mathrm{d}\sigma = D(\theta)\,\mathrm{d}\Omega$$

with D being the *differential cross section*. We have

$$\begin{aligned} D = \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \left|\frac{b}{\sin\theta}\frac{\mathrm{d}b}{\mathrm{d}\theta}\right| \\ &\overset{\text{hard-sphere}}{=} \frac{R^2}{4} \\ &\overset{\text{Rutherford}}{=} \left(\frac{q_1 q_2}{4E\sin^2(\theta/2)}\right)^2 \end{aligned}$$

The cross section can then be found as

$$\begin{aligned} \sigma &= \int \mathrm{d}\sigma \\ &\overset{\text{hard-sphere}}{=} \pi R^2 \sim \text{circle} \\ &\overset{\text{Rutherford}}{=} \infty \sim \text{infinite range} \end{aligned}$$

Any particles within this area will scatter while any outside will pass by unaffected.

Consider a beam of particles with uniform *luminosity* L. Then $\mathrm{d}N = L\,\mathrm{d}\sigma$ is the number of particles per unit time passing through the area $\mathrm{d}\sigma$ implying

$$\mathrm{d}N = L\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\,\mathrm{d}\Omega$$

or if L and $\mathrm{d}\Omega$ are fixed then

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\mathrm{d}N}{L\,\mathrm{d}\Omega}$$

6.2. Fermi's *Golden Rule*

Consider a particle decaying as

$$1 \to 2 + 3 + \ldots + n$$

Then the decay rate is given by

$$\Gamma = \frac{S}{2\hbar m_1}\int |\mathscr{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 - p_2 - \ldots - p_n)$$

$$\times \prod_{j=2}^{n} 2\pi\delta\left(p_j^2 - m_j^2c^2\right)\Theta\left(p_j^0\right)\frac{\mathrm{d}^4 p_j}{(2\pi)^4}$$

where S is the *symmetry factor*, for each group of s identical particles S gets a factor of $(1/s!)$. The dynamics of the process are contained within the *amplitude* $\mathscr{M}(p_1, \ldots, p_n)$ everything else is referred to as *phase space*.[20] We are told to integrate over all outgoing four-momenta with the constraints:

1. All outgoing particles lie on their mass shell $p_j^2 = m_j^2c^2$.
2. All outgoing energies are positive $p_j^0 > 0$.
3. Energy and momentum are conserved $p_1 = p_2 + \ldots + p_n$.

These are enforced by the δ and Θ functions.

We can write $\mathrm{d}^4 p_j = \mathrm{d}p_j^0\,\mathrm{d}^3\boldsymbol{p}_j$ and

$$\delta\left(p_j^2 - m_j^2c^2\right) = \delta\left(\left(p_j^0\right)^2 - \left(\boldsymbol{p}_j^2 + m_j^2c^2\right)\right)$$

[20]The units of $\mathscr{M}$ are $(mc)^{4-n}$ where $n = \#$external lines.

$$= \frac{1}{2\sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}} \left[\delta\left(p_j^0 - \sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}\right) + \delta\left(p_j^0 + \sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}\right)\right]$$

The Θ function kills the spike at $p_j^0 = -\sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}$ implying

$$\Theta\left(p_j^0\right)\delta\left(p_j^2 - m_j^2 c^2\right) = \frac{1}{2\sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}}\delta\left(p_j^0 - \sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}\right)$$

We can now perform the integral over p_j^0 using the δ function and obtain[21]

$$\Gamma = \frac{S}{2\hbar m_1} \int |\mathscr{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 - \ldots - p_n) \prod_{j=2}^{n} \frac{1}{2\sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}} \frac{\mathrm{d}^3 \boldsymbol{p}_j}{(2\pi)^3}$$

with

$$p_j^0 \to \sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}$$

everywhere.

As an example consider the decay

$$1 \to 2 + 3$$

Then

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta^{(4)}(p_1 - p_2 - p_3)}{\sqrt{\boldsymbol{p}_2^2 + m_2^2 c^2}\sqrt{\boldsymbol{p}_3^2 + m_3^2 c^2}} \, \mathrm{d}^3\boldsymbol{p}_2 \, \mathrm{d}^3\boldsymbol{p}_3$$

Also,

$$\delta^{(4)}(p_1 - p_2 - p_3) = \delta(p_1^0 - p_2^0 - p_3^0)\delta^{(3)}(\boldsymbol{p}_1 - \boldsymbol{p}_2 - \boldsymbol{p}_3)$$

We assume $\boldsymbol{p}_1 = 0$ and $p_1^0 = m_1 c$

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta\left(m_1 c - \sqrt{\boldsymbol{p}_2^2 + m_2^2 c^2} - \sqrt{\boldsymbol{p}_3^2 + m_3^2 c^2}\right)}{\sqrt{\boldsymbol{p}_2^2 + m_2^2 c^2}\sqrt{\boldsymbol{p}_3^2 + m_3^2 c^2}} \delta^{(3)}(\boldsymbol{p}_2 + \boldsymbol{p}_3) \, \mathrm{d}^3\boldsymbol{p}_2 \, \mathrm{d}^3\boldsymbol{p}_3$$

$$= \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta\left(m_1 c - \sqrt{\boldsymbol{p}_2^2 + m_2^2 c^2} - \sqrt{\boldsymbol{p}_2^2 + m_3^2 c^2}\right)}{\sqrt{\boldsymbol{p}_2^2 + m_2^2 c^2}\sqrt{\boldsymbol{p}_2^2 + m_3^2 c^2}} \, \mathrm{d}^3\boldsymbol{p}_2$$

We adopt spherical coordinates with $\mathrm{d}^3\boldsymbol{p}_2 = r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\varphi$[22]

[21]Since Θ is unity at $p_j^0 = \sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}$

[22]This is momentum space $r = |\boldsymbol{p}_2|$.

$$\Gamma = \frac{S}{32\pi^2\hbar m_1}\int |\mathscr{M}|^2 \frac{\delta\left(m_1c - \sqrt{r^2+m_2^2c^2} - \sqrt{r^2+m_3^2c^2}\right)}{\sqrt{r^2+m_2^2c^2}\sqrt{r^2+m_3^2c^2}} r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\varphi$$

We had $\mathscr{M} = \mathscr{M}(p_1, p_2, p_3)$ initially and after everything we now have $\mathscr{M} = \mathscr{M}(\boldsymbol{p}_2)$. However, $\mathscr{M}$ is a scalar implying $\mathscr{M} = \mathscr{M}(\boldsymbol{p}_2^2 = r^2)$. Then

$$\Gamma = \frac{S}{8\pi\hbar m_1}\int_0^\infty |\mathscr{M}(r)|^2 \frac{\delta\left(m_1c - \sqrt{r^2+m_2^2c^2} - \sqrt{r^2+m_3^2c^2}\right)}{\sqrt{r^2+m_2^2c^2}\sqrt{r^2+m_3^2c^2}} r^2 \, \mathrm{d}r$$

We define

$$u = \sqrt{r^2+m_2^2c^2} + \sqrt{r^2+m_3^2c^2}$$

Then

$$\Gamma = \frac{S}{8\pi\hbar m_1}\int_{(m_2+m_3)c}^\infty |\mathscr{M}(r)|^2 \delta(m_1c - u)\frac{r}{u} \, \mathrm{d}u$$

Using the δ function we send $u \to m_1c$ and r to

$$r_0 = \frac{c}{2m_1}\sqrt{m_1^4 + m_2^4 + m_3^4 - 2m_1^2m_2^2 - 2m_1^2m_3^2 - 2m_2^2m_3^2}$$

which is the value of $|\boldsymbol{p}_2|$ consistent with conservation of energy. Then[23]

$$\Gamma = \frac{S|\boldsymbol{p}|}{8\pi\hbar m_1^2 c}|\mathscr{M}|^2$$

where $|\boldsymbol{p}|$ is the magnitude of either outgoing momentum.

6.3. Scattering

Consider the process

$$1 + 2 \to 3 + 4 + \ldots + n$$

Then the cross section is given by

$$\sigma = \frac{S\hbar^2}{4\sqrt{(p_1\cdot p_2)^2 - (m_1m_2c^2)^2}}\int |\mathscr{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - \ldots - p_n)$$

[23]Being able to perform all the integrals as above without knowing the form of $\mathscr{M}$ is typically not possible. Also, we assume $m_1 > m_2 + m_3$ otherwise $\Gamma = 0$ as expected.

$$\times \prod_{j=3}^{n} 2\pi\delta\big(p_j^2 - m_j^2c^2\big)\Theta\Big(p_j^0\Big)\frac{\mathrm{d}^4 p_j}{(2\pi)^4}$$

We see everything is almost the same as before and we can again perform the p_j^0 integrals

$$\sigma = \frac{S\hbar^2}{4\sqrt{\left(p_1 \cdot p_2\right)^2 - \left(m_1 m_2 c^2\right)^2}} \int |\mathscr{M}|^2 (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - \ldots - p_n)$$

$$\times \prod_{j=3}^{n} \frac{1}{2\sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}} \frac{\mathrm{d}^3 \boldsymbol{p}_j}{(2\pi)^3}$$

with

$$p_j^0 = \sqrt{\boldsymbol{p}_j^2 + m_j^2 c^2}$$

everywhere.

As an example consider the process

$$1 + 2 \to 3 + 4$$

in the CM frame $\boldsymbol{p}_2 = -\boldsymbol{p}_1$ and

$$\sqrt{\left(p_1 \cdot p_2\right)^2 - \left(m_1 m_2 c^2\right)^2} = \frac{(E_1 + E_2)|\boldsymbol{p}_1|}{c}$$

Then

$$\sigma = \frac{S\hbar^2 c}{64\pi^2 (E_1 + E_2)|\boldsymbol{p}_1|} \int |\mathscr{M}|^2 \frac{\delta^{(4)}(p_1 + p_2 - p_3 - p_4)}{\sqrt{\boldsymbol{p}_3^2 + m_3^2 c^2}\sqrt{\boldsymbol{p}_4^2 + m_4^2 c^2}} \,\mathrm{d}^3\boldsymbol{p}_3 \,\mathrm{d}^3\boldsymbol{p}_4$$

We write

$$\delta^{(4)}(p_1 + p_2 - p_3 - p_4) = \delta\left(\frac{E_1 + E_2}{c} - p_3^0 - p_4^0\right)\delta^{(3)}(\boldsymbol{p}_3 + \boldsymbol{p}_4)$$

and perform the $\boldsymbol{p}_4$ integral using the δ function

$$\sigma = \left(\frac{\hbar}{8\pi}\right)^2 \frac{Sc}{(E_1 + E_2)|\boldsymbol{p}_1|} \int |\mathscr{M}|^2 \frac{\delta\Big((E_1 + E_2)/c - \sqrt{\boldsymbol{p}_3^2 + m_3^2 c^2} - \sqrt{\boldsymbol{p}_3^2 + m_4^2 c^2}\Big)}{\sqrt{\boldsymbol{p}_3^2 + m_3^2 c^2}\sqrt{\boldsymbol{p}_3^2 + m_4^2 c^2}} \,\mathrm{d}^3\boldsymbol{p}_3$$

This time we cannot do the angular integrals since $\mathscr{M}$ depends on the direction of $\boldsymbol{p}_3$ as well as its magnitude.[24] We still write

$$\mathrm{d}^3\boldsymbol{p}_3 = r^2\,\mathrm{d}r\,\mathrm{d}\Omega$$

Then

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\hbar}{8\pi}\right)^2 \frac{Sc}{(E_1+E_2)|\boldsymbol{p}_1|} \underbrace{\int_0^\infty |\mathscr{M}|^2 \frac{\delta\Big((E_1+E_2)/c - \sqrt{r^2+m_3^2c^2} - \sqrt{r^2+m_4^2c^2}\Big)}{\sqrt{r^2+m_3^2c^2}\sqrt{r^2+m_4^2c^2}} r^2\,\mathrm{d}r}_{\text{as before}}$$

We do the integral as before to obtain

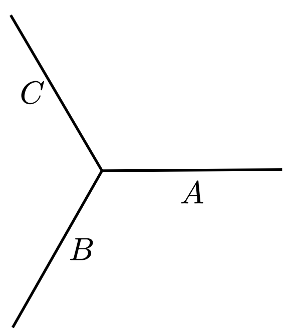
$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathscr{M}|^2}{(E_1+E_2)^2} \frac{|\boldsymbol{p}_f|}{|\boldsymbol{p}_i|}$$

where $|\boldsymbol{p}_f|$ and $|\boldsymbol{p}_i|$ are the magnitudes of either- outgoing and incoming momentum respectively.

6.4. ABC theory

We now want to begin computing ampltitudes $\mathscr{M}$. This is done by writing all the relevant Feynman diagrams for a given process and then using the *Feynman rules.* We start by considering a simple *toy* theory which will serve to illustrate the method.

We imagine a world with three particles $\{A, B, C\}$.[25] We assume they are spin zero and each is its own antiparticle. There is one primitive vertex

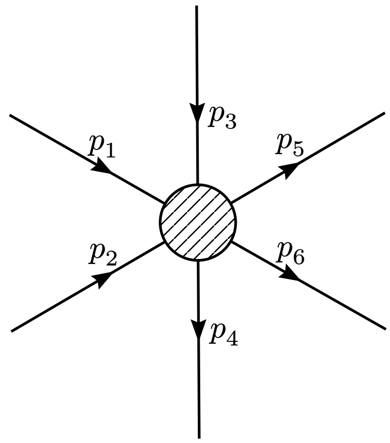


This diagram contributes to $A \to B + C$. We will also consider other processes such as $A + A \to B + B$ and $A + B \to A + B$ both of which happen through the exchange of a C.

[24] We have three invariants $\boldsymbol{p}_1^2$, $\boldsymbol{p}_3^2$ and $\boldsymbol{p}_1 \cdot \boldsymbol{p}_3 = |\boldsymbol{p}_1||\boldsymbol{p}_3|\cos\theta$. However, $\boldsymbol{p}_1$ is fixed so the only integration variables $\mathscr{M}$ can depend on are $|\boldsymbol{p}_3|$ and θ.

[25] We will assume $m_A > m_B + m_C$.

We now consider a general Feynman diagram



Then the Feynman rules are:

1. Label the incoming- and outgoing momenta $p_1, ..., p_n$ as above and label internal momenta $q_1, ...$
2. Each vertex carries a factor $-ig$. Where g is referred to as the *coupling*.
3. Each internal line carries a *propagator*[26]
$$\frac{i}{q_j^2 - m_j^2 c^2}$$
4. Each vertex carries a δ function
$$(2\pi)^4 \delta^{(4)}(k_1 + k_2 + k_3)$$
with k_i being negative for outward momenta.
5. Each internal line carries an integral
$$\int \frac{\mathrm{d}^4 q_j}{(2\pi)^4}$$
6. The result will carry a factor
$$(2\pi)^4 \delta^{(4)}(p_1 + p_2 + ... - p_n)$$
replace this by i.

As an example we consider the decay

$$A \to B + C$$

[26]We have $q_j^2 \neq m_j^2 c^2$ since *virtual particles* are off shell.

This diagram has no internal lines and we obtain

$$(-ig)(2\pi)^4\delta^{(4)}(p_1 - p_2 - p_3)$$

which upon $(2\pi)^4\delta^{(4)} \to i$ becomes

$$\mathscr{M} = g + \mathcal{O}(g^2)$$

Then

$$\Gamma = \frac{g^2|\boldsymbol{p}|}{8\pi\hbar m_A^2 c}$$

where

$$|\boldsymbol{p}| = \frac{c}{2m_A}\sqrt{m_A^4 + m_B^4 + m_C^4 - 2m_A^2 m_B^2 - 2m_A^2 m_C^2 - 2m_B^2 m_C^2}$$

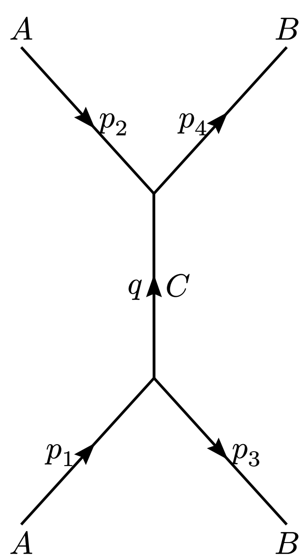
We have found the lifetime of A

$$\tau_A = \frac{8\pi\hbar m_A^2 c}{g^2|\boldsymbol{p}|}$$

As a second example we consider the process

$$A + A \to B + B$$

The lowest-order contribution is



We obtain

$$(-ig)^2 \int \frac{\mathrm{d}^4 q}{(2\pi)^4}\frac{i}{q^2 - m_C^2 c^2}(2\pi)^4\delta^{(4)}(p_1 - q - p_3)(2\pi)^4\delta^{(4)}(p_2 + q - p_4)$$

$$-ig^2(2\pi)^4 \frac{\delta^{(4)}(p_1 + p_2 - p_3 - p_4)}{(p_4 - p_2)^2 - m_C^2 c^2}$$

which upon $(2\pi)^4 \delta^{(4)} \to i$ becomes

$$\mathscr{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2}$$

There is a second diagram of order $\mathcal{O}(g^2)$ obtained by *twisting* the B lines. This amounts to $p_3 \leftrightarrow p_4$ meaning the total amplitude is

$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

We now find $\mathrm{d}\sigma/\mathrm{d}\Omega$ in the CM frame. We assume $m_A = m_B = m$ and $m_C = 0$ then

$$(p_4 - p_2)^2 - m_C^2 c^2 = p_4^2 + p_2^2 - 2p_2 \cdot p_4 = -2\boldsymbol{p}^2(1 - \cos\theta)$$

$$(p_3 - p_2)^2 - m_C^2 c^2 = p_3^2 + p_2^2 - 2p_3 \cdot p_2 = -2\boldsymbol{p}^2(1 + \cos\theta)$$

with $\boldsymbol{p}$ being the incoming momentum of particle 1. Then

$$\mathscr{M} = -\frac{g^2}{\boldsymbol{p}^2 \sin^2\theta}$$

and

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{2}\left(\frac{\hbar c g^2}{16\pi E \boldsymbol{p}^2 \sin^2\theta}\right)^2$$

since $S = 1/2$.

7. Quantum Electrodynamics

The ABC theory described above is a perfectly legitimate QFT. However, it does not describe the real world. This is partly due to particles being described differently depending on their spin in real QFTs:

$$\text{spin } 0 \sim \text{Klein-Gordon equation}$$

$$\text{spin } 1/2 \sim \text{Dirac equation}$$

$$\text{spin } 1 \sim \text{Proca equation}$$

Once we have established the Feynman rules, however, then the underlying field equation *becomes obsolete*. We would like to understand quantum electrodynamics meaning we need to understand the dynamics of e^- and γ.

7.1. The Dirac equation

We can "derive" the Klein-Gordon equation by considering

$$p^\mu p_\mu - m^2c^2 = 0$$

and promoting p_μ to an operator

$$p_\mu \to i\hbar\partial_\mu$$

Then we obtain

$$\partial^\mu\partial_\mu\psi + \frac{m^2c^2}{\hbar^2}\psi = 0$$

or with $\Box \equiv \partial_\mu\partial^\mu$ we have

$$\left[\Box + \frac{m^2c^2}{\hbar^2}\right]\psi = 0$$

which is the Klein-Gordon equation. A more correct derivation uses the Klein-Gordon Lagrangian

$$\mathcal{L}_{\text{KG}} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2$$

which describes a free massive scalar field ϕ.

When first conceived the Klein-Gordon equation appeared very problematic.[27] This lead Dirac to seek an equation consistent with

$$p^\mu p_\mu - m^2c^2 = 0$$

while also being first order in time. The idea is to "factor" the above

$$p^\mu p_\mu - m^2c^2 = (\beta^\kappa p_\kappa + mc)\big(\gamma^\lambda p_\lambda - mc\big)$$

$$= \beta^\kappa \gamma^\lambda p_\kappa p_\lambda - mc(\beta^\kappa - \gamma^\kappa)p_\kappa - m^2c^2$$

Then we need $\beta^\kappa = \gamma^\kappa$ and[28]

$$p^\mu p_\mu = \gamma^\kappa \gamma^\lambda p_\kappa p_\lambda$$

$$= \frac{1}{2}\big(\gamma^\kappa\gamma^\lambda + \gamma^\lambda\gamma^\kappa\big)p_\kappa p_\lambda$$

$$\stackrel{!}{=} \eta^{\kappa\lambda} p_\kappa p_\lambda$$

which implies the γ^μ are matrices satisfying the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$$

We will use the 4×4 matrices

$$\gamma^0 = \begin{pmatrix} \mathbb{1}_2 & 0 \\ 0 & -\mathbb{1}_2 \end{pmatrix}; \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

However, many other γ^μ also do the job. Then

$$\gamma^\mu p_\mu - mc = 0$$

and promoting p_μ to an operator we obtain

$$\big(i\hbar\gamma^\mu \partial_\mu - mc\big)\psi = 0$$

which is the Dirac equation. We will call ψ the Dirac spinor. Note, the components of ψ do not transform like a four-vector!

[27] When trying to use it to describe e^- people ran into problems. This is due to the Klein-Gordon equation describing particles with spin 0. The Klein-Gordon equation is also second order in time leading to an apparent breaking of Born's statistical interpretation.

[28] Since $p_\kappa p_\lambda$ is symmetric.

7.2. Solutions to the Dirac equation

Consider $\psi(\boldsymbol{x}, t) = \psi(t)$.[29] Then the Dirac equation becomes

$$i\hbar\gamma^0\partial_t\psi - mc^2\psi = 0$$

or

$$\begin{pmatrix} \mathbb{1}_2 & 0 \\ 0 & -\mathbb{1}_2 \end{pmatrix} \begin{pmatrix} \partial_t\psi_A \\ \partial_t\psi_B \end{pmatrix} = -i\frac{mc^2}{\hbar}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}$$

implying

$$\partial_t\psi_A = -i\left(\frac{mc^2}{\hbar}\right)\psi_A$$

$$\partial_t\psi_B = i\left(\frac{mc^2}{\hbar}\right)\psi_B$$

with solutions

$$\psi_A = \psi_A(0)\exp\left[-i\left(\frac{mc^2}{\hbar}\right)t\right]$$

$$\psi_B = \psi_B(0)\exp\left[+i\left(\frac{mc^2}{\hbar}\right)t\right]$$

We recognize the familiar

$$\text{time dependence} \sim \exp\left(-\frac{iEt}{\hbar}\right)$$

Then ψ_A corresponds to a particle with $E = mc^2$ as expected for a particle at rest. However, ψ_B corresponds to a particle with negative energy $E = -mc^2$! We take these negative energy solutions to represent antiparticles with positve energy. Then ψ_A describes e^- while ψ_B describes e^+. Both being two-component spinors which can nicely describe a spin $\frac{1}{2}$ system. We have found four independent solutions with $\boldsymbol{p} = 0$

[29] Then the Dirac equation describes a state with $\boldsymbol{p} = 0$ i.e. a particle at rest.

$$\psi^{(1)} = \exp\left[-i\left(\frac{mc^2}{\hbar}\right)t\right]\begin{pmatrix}1\\0\\0\\0\end{pmatrix} \sim \text{spin-up } e^-$$

$$\psi^{(2)} = \exp\left[-i\left(\frac{mc^2}{\hbar}\right)t\right]\begin{pmatrix}0\\1\\0\\0\end{pmatrix} \sim \text{spin-down } e^-$$

$$\psi^{(3)} = \exp\left[+i\left(\frac{mc^2}{\hbar}\right)t\right]\begin{pmatrix}0\\0\\1\\0\end{pmatrix} \sim \text{spin-down } e^+$$

$$\psi^{(4)} = \exp\left[+i\left(\frac{mc^2}{\hbar}\right)t\right]\begin{pmatrix}0\\0\\0\\1\end{pmatrix} \sim \text{spin-up } e^+$$

Consider plane-wave solutions

$$\psi(x) = ae^{-ik^\mu x_\mu}u(k)$$

implying

$$\partial_\mu \psi = -ik_\mu \psi$$

so

$$\left(\hbar\gamma^\mu k_\mu - mc\right)u = 0$$

Using

$$\gamma^\mu k_\mu = \begin{pmatrix} k^0 & -\boldsymbol{k}\cdot\boldsymbol{\sigma} \\ \boldsymbol{k}\cdot\boldsymbol{\sigma} & -k^0 \end{pmatrix}$$

we find

$$u_A = \frac{1}{k^0 - mc/\hbar}(\boldsymbol{k}\cdot\boldsymbol{\sigma})u_B$$

$$u_B = \frac{1}{k^0 + mc/\hbar}(\boldsymbol{k}\cdot\boldsymbol{\sigma})u_A$$

or

$$u_A = \frac{1}{(k^0)^2 - (mc/\hbar)^2}(\boldsymbol{k}\cdot\boldsymbol{\sigma})^2 u_A$$

$$\overset{(\boldsymbol{k}\cdot\boldsymbol{\sigma})^2=\boldsymbol{k}^2}{=} \frac{\boldsymbol{k}^2}{(k^0)^2 - (mc/\hbar)^2} u_A$$

implying[30]

$$k^\mu k_\mu = \left(\frac{mc}{\hbar}\right)^2$$

Then $\hbar k^\mu$ must be a four-vector whose square is m^2c^2 implying

$$\hbar k^\mu = \pm p^\mu$$

with $+$ being associated with particle states and $-$ being associated with antiparticle states. We can now construct our four independent solutions:

$$u_A = \begin{pmatrix}1\\0\end{pmatrix} \Rightarrow u_B = \frac{\boldsymbol{p}\cdot\boldsymbol{\sigma}}{p^0 + mc}\begin{pmatrix}1\\0\end{pmatrix} = \frac{c}{E + mc^2}\begin{pmatrix}p_z\\p_z + ip_y\end{pmatrix}$$

$$u_A = \begin{pmatrix}0\\1\end{pmatrix} \Rightarrow u_B = \frac{c}{E + mc^2}\begin{pmatrix}p_x - ip_y\\-p_z\end{pmatrix}$$

$$u_B = \begin{pmatrix}1\\0\end{pmatrix} \Rightarrow u_A = \frac{c}{E + mc^2}\begin{pmatrix}p_z\\p_x + ip_y\end{pmatrix}$$

$$u_B = \begin{pmatrix}0\\1\end{pmatrix} \Rightarrow u_A = \frac{c}{E + mc^2}\begin{pmatrix}p_x - ip_y\\-p_z\end{pmatrix}$$

We need the $+$ sign in the first two, otherwise the u_B would blow up as $\boldsymbol{p} \to 0$. These are again our particle states. We will normalize these by

$$u^\dagger u = \frac{2E}{c}$$

We find the normalisation

[30]or the trivial solution $u_A = u_B = 0$

$$N \equiv \sqrt{\frac{E+mc^2}{c}}$$

and the canonical solutions become

$$u^{(1)} = N \begin{pmatrix} 1 \\ 0 \\ \frac{cp_z}{E+mc^2} \\ \frac{c(p_x+ip_y)}{E+mc^2} \end{pmatrix}; \quad u^{(2)} = N \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_x-ip_y)}{E+mc^2} \\ \frac{-cp_z}{E+mc^2} \end{pmatrix}$$

$$v^{(1)} = N \begin{pmatrix} \frac{c(p_x-ip_y)}{E+mc^2} \\ \frac{-cp_z}{E+mc^2} \\ 0 \\ 1 \end{pmatrix}; \quad v^{(2)} = -N \begin{pmatrix} \frac{cp_z}{E+mc^2} \\ \frac{c(p_x+ip_y)}{E+mc^2} \\ 1 \\ 0 \end{pmatrix}$$

with

$$\psi = a \exp\left[-\frac{ip_\mu x^\mu}{\hbar}\right] u \sim \text{particles}$$

$$\psi = a \exp\left[+\frac{ip_\mu x^\mu}{\hbar}\right] v \sim \text{antiparticles}$$

The particle states satisfy

$$(\gamma^\mu p_\mu - mc)u = 0$$

However, the antiparticle states satisfy

$$(\gamma^\mu p_\mu + mc)v = 0$$

7.3. Bilinear covariants

We would like to construct a scalar from ψ. This is done by introducing the adjoint $\overline{\psi}$ defined by

$$\overline{\psi} \equiv \psi^\dagger \gamma^0$$

Then

$$\overline{\psi}\psi = \psi^\dagger \gamma^0 \psi \sim \text{scalar}$$

We can also construct a pseudoscalar by introducing γ^5

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$$

with

$$\{\gamma^\mu, \gamma^5\} = 0$$

Then

$$\overline{\psi}\gamma^5\psi \sim \text{pseudoscalar}$$

Likewise,

$$\overline{\psi}\gamma^\mu\psi \sim \text{vector}$$

$$\overline{\psi}\gamma^\mu\gamma^5\psi \sim \text{pseudovector}$$

$$\overline{\psi}\sigma^{\mu\nu}\psi \sim \text{antisymmetric tensor}$$

with

$$\sigma^{\mu\nu} \equiv \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\mu\gamma^\nu)$$

Any other *biliniear* will always reduce to one of the above. One can further show that

$$j^\mu = \overline{\psi}\gamma^\mu\psi$$

is conserved and leads to charge!

7.4. γ and Maxwell's equations

We define the four-potential

$$A^\mu = (V, \boldsymbol{A})$$

and the four-current

$$J^\mu = (c\rho, \boldsymbol{J})$$

Then Maxwell's equations become[31]

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c}J^\nu$$

where we have introduced the antisymmetric Faraday tensor

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

Note, this implies

[31] The inhomogeneous ones.

$$\partial_\mu J^\mu = 0$$

We see Maxwell's equations are unchanged under gauge transformations

$$A_\mu \to A_\mu + \partial_\mu \lambda$$

We refer to this as having gauge freedom. We can use this to impose

$$\partial_\mu A^\mu = 0$$

which is called the Lorentz gauge. With this gauge Maxwell's equations become

$$\Box A^\mu = \frac{4\pi}{c} J^\mu$$

This still does not uniquely specify A^μ since any gauge transformation with

$$\Box \lambda = 0$$

will still leave our equations unchanged. We will remedy this by imposing

$$A^0 = 0$$

in empty space where $J^\mu = 0$. Then the Lorentz gauge becomes

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0$$

which is called the Coulomb gauge. Whenever we single out a component like A^0 we break the otherwise manifest Lorentz invariance. Then whenever we perform a Lorentz transformation we are forced to also perform a gauge transformation in order to restore the Coulomb gauge.

A^μ becomes the wave function of γ in quantum electrodynamics. The free γ satisfies the above with $J^\mu = 0$

$$\Box A^\mu = 0$$

which we see is the Klein-Gordon equation for a particle with $m = 0$. Consider plane-wave solutions

$$A^\mu(x) = a \exp\left[-\frac{i p_\mu x^\mu}{\hbar}\right] \epsilon^\mu(p)$$

where $p^\mu = (E/c, \boldsymbol{p})$ and ϵ^μ is the *polarization vector*.[32] We find the constraint

[32] ϵ^μ characterises the spin of γ.

$$p^\mu p_\mu = 0$$

as we would expect. We also have by the Lorentz gauge

$$p^\mu \epsilon_\mu = 0$$

and in the Coulomb gauge

$$\epsilon^0 = 0$$

implying

$$\boldsymbol{\epsilon} \cdot \boldsymbol{p} = 0$$

Then $\boldsymbol{\epsilon}$ is perpendicular to the direction of propagation. We say a free γ is *transversely polarized*. There are always two linearly independent $\boldsymbol{\epsilon}$ perpendicular to $\boldsymbol{p}$. As an example if $\boldsymbol{p}$ is along $\hat{z}$ then

$$\boldsymbol{\epsilon}^{(1)} = (1, 0, 0); \quad \boldsymbol{\epsilon}^{(2)} = (0, 1, 0)$$

We would expect a γ has three spin states. However, only massive particles with spin s admit $2s + 1$ different spin orientations. All massless particles only have two.[33] Then along the direction of motion the γ can have

$$m_s = \pm s \sim h = \pm 1$$

7.5. Feynman rules for QED

We briefly summarize what we have found above.

We found that free e^- and e^+ with $p = (E/c, \boldsymbol{p})$ are represented by

$$\psi(x) = a \exp\left[-\frac{ip_\mu x^\mu}{\hbar}\right] u^{(s)}(p) \sim e^-$$

$$\psi(x) = a \exp\left[+\frac{ip_\mu x^\mu}{\hbar}\right] v^{(s)}(p) \sim e^+$$

with $s = 1, 2$. The spinors $u^{(s)}$ and $v^{(s)}$ satisfy

$$(\gamma^\mu p_\mu - mc)u = 0; \quad (\gamma^\mu p_\mu + mc)v = 0$$

and their adjoints satisfy

[33] expect for $s = 0$ which only has one.

$$\overline{u}(\gamma^\mu p_\mu - mc) = 0; \quad \overline{v}(\gamma^\mu p_\mu + mc) = 0$$

These are orthogonal

$$\overline{u}^{(1)} u^{(2)} = 0; \quad \overline{v}^{(1)} v^{(2)} = 0$$

normalised

$$\overline{u}u = 2mc; \quad \overline{v}v = -2mc$$

and complete

$$\sum u^{(s)} \overline{u}^{(s)} = \gamma^\mu p_\mu + mc; \quad \sum v^{(s)} \overline{v}^{(s)} = \gamma^\mu p_\mu - mc$$

We have already determined a convenient set $\{u^{(i)}, v^{(i)}\}$.

We also found that free γ with $p = (E/c, \boldsymbol{p})$ are represented by

$$A_\mu(x) = a \exp\left[-\frac{i p_\mu x^\mu}{\hbar}\right] \epsilon_\mu^{(s)}$$

where $s = 1, 2$. The $\epsilon_\mu^{(s)}$ satisfy

$$p^\mu \epsilon_\mu = 0$$

are orthogonal

$$\epsilon_\mu^{(1)*} \epsilon^{(2)\mu} = 0$$

and normalised[34]

$$\epsilon^{\mu *} \epsilon_\mu = 1$$

The Coulomb gauge has

$$\epsilon^0 = 0; \quad \boldsymbol{\epsilon} \cdot \boldsymbol{p} = 0$$

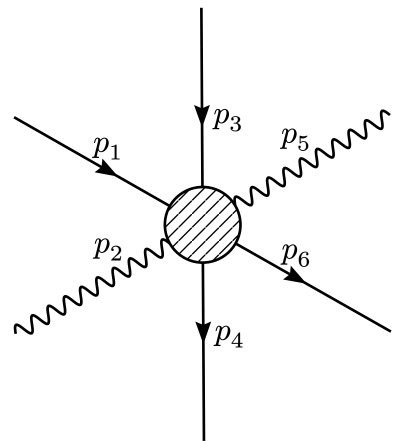
and the $\boldsymbol{\epsilon}$ are complete

$$\sum \epsilon_i^{(s)} \epsilon_j^{(s)*} = \delta_{ij} - \hat{p}_i \hat{p}_j$$

We have already determined a convenient pair $\{\epsilon^{(1)}, \epsilon^{(2)}\}$.

We now consider a general Feynman diagram in QED

[34] Note, these are arbitrary choices.



Then the Feynman rules are:

1. Label the incoming- and outgoing momenta $p_1, ..., p_n$ as above and label internal momenta $q_1, ...$[35]

2. Each external line carries a factor:

$$e^- \sim \begin{cases} \text{incoming: } u \\ \text{outgoing: } \overline{u} \end{cases}$$

$$e^+ \sim \begin{cases} \text{incoming: } \overline{v} \\ \text{outgoing: } v \end{cases}$$

$$\gamma \sim \begin{cases} \text{incoming: } \epsilon_\mu \\ \text{outgoing: } \epsilon^*_\mu \end{cases}$$

3. Each vertex carries a factor $ig_e\gamma^\mu$. Where g_e is related to the fundamental charge e by

$$g_e = e\sqrt{\frac{4\pi}{\hbar c}} = \sqrt{4\pi\alpha}$$

4. Each internal line carries a propagator:

$$e^\pm \sim \frac{i\gamma^\mu q_\mu + mc}{q^2 - m^2c^2}$$

$$\gamma \sim \frac{-i\eta_{\mu\nu}}{q^2}$$

[35]Note, the external momenta should go forward in time.

5. Each vertex carries a δ function

$$(2\pi)^4\delta^{(4)}(k_1 + k_2 + k_3)$$

with k_i being negative for outward momenta.

6. Each internal line carries an integral

$$\int \frac{\mathrm{d}^4 q}{(2\pi)^4}$$

7. The result will carry a factor

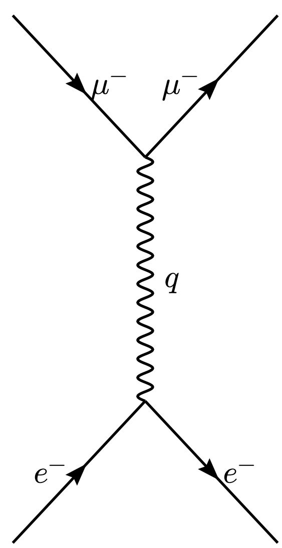
$$(2\pi)^4\delta^{(4)}(p_1 + p_2 + \ldots - p_n)$$

replace this by i.

8. We include a $-$ between diagrams differing only by the interchange of two incoming or outgoing $e^{\pm}$[36]

7.6. $e^-\mu^-$ scattering

Consider $e^-\mu^-$ scattering. We have one diagram contributing to second-order



with q upwards. We find

$$\mathscr{M} \sim \int \frac{\mathrm{d}^4 q}{(2\pi)^4} \overbrace{\left[\overline{u}^{(s_3)}(p_3) i g_e \gamma^\mu u^{(s_1)}(p_1)\right]}^{e^- \text{ line}} \frac{-i\eta_{\mu\nu}}{q^2} \underbrace{\left[\overline{u}^{(s_4)}(p_4) i g_e \gamma^\nu u^{(s_2)}(p_2)\right]}_{\mu^- \text{ line}}$$

[36]or of an incoming $e^{\mp}$ and outgoing $e^{\pm}$

$$\times (2\pi)^4 \delta^{(4)}(p_2 - p_4 + q)(2\pi)^4 \delta^{(4)}(p_1 - p_3 - q)$$

Above we have constructed the "e^- line" and "μ^- line" *backwards* and *connected* them with the γ propagator. We do this to ensure the matrix multiplication is consistent. Then

$$\mathscr{M} = \left[\overline{u}^{(s_3)}(p_3) i g_e \gamma^\mu u^{(s_1)}(p_1)\right] \frac{\eta_{\mu\nu}}{(p_1 - p_3)^2} \left[\overline{u}^{(s_4)}(p_4) i g_e \gamma^\nu u^{(s_2)}(p_2)\right]$$

$$= -\frac{g_e^2}{(p_1 - p_3)^2} \left[\overline{u}^{(s_3)}(p_3) \gamma^\mu u^{(s_1)}(p_1)\right] \left[\overline{u}^{(s_4)}(p_4) \gamma_\mu u^{(s_2)}(p_2)\right]$$

7.7. e^-e^- scattering

We have the same diagram as above. However, now we also have a second diagram where the e^- emerging with (p_3, s_3) comes from the e^- with (p_2, s_2) instead of the e^- with (p_1, s_1). We essentially twist the diagram above. The total amplitude is then

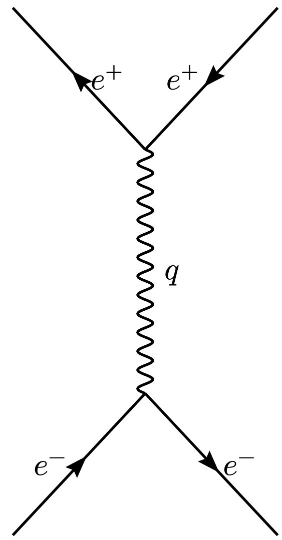
$$\mathscr{M} = -\frac{g_e^2}{(p_1 - p_3)^2} \left[\overline{u}(3) \gamma^\mu u(1)\right] \left[\overline{u}(4) \gamma_\mu u(2)\right]$$

$$+ \underbrace{\frac{g_e^2}{(p_1 - p_4)^2} \left[\overline{u}(4) \gamma^\mu u(1)\right] \left[\overline{u}(3) \gamma_\mu u(2)\right]}_{3 \leftrightarrow 4}$$

where $-$ comes from rule 8.

7.8. e^-e^+ scattering

We again have two diagrams. One is similar to $e^-\mu^-$ scattering



We find

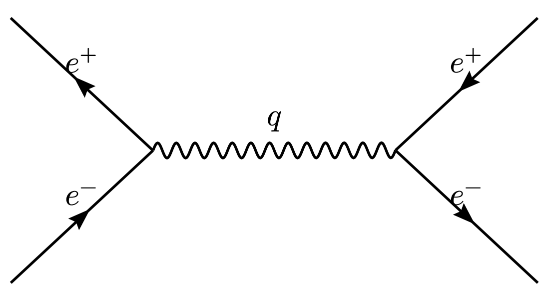
$$\mathscr{M}_1 \sim \int \frac{\mathrm{d}^4 q}{(2\pi)^4} [\overline{u}(3) i g_e \gamma^\mu u(1)] \frac{-i\eta_{\mu\nu}}{q^2} \underbrace{[\overline{v}(2) i g_e \gamma^\nu v(4)]}_{e^+ \text{ line}}$$

$$\times (2\pi)^4 \delta^{(4)}(p_1 - p_3 - q)(2\pi)^4 \delta^{(4)}(p_2 + q - p_4)$$

Note, when going *backwards* along an antiparticle we are going forwards in time.[37] Then

$$\mathscr{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2} [\overline{u}(3)\gamma^\mu u(1)] \left[\overline{v}(2)\gamma_\mu v(4)\right]$$

The second diagram is



We find

$$\mathscr{M}_2 \sim \int \frac{\mathrm{d}^4 q}{(2\pi)^4} [\overline{v}(2) i g_e \gamma^\mu u(1)] \frac{-i\eta_{\mu\nu}}{q^2} [\overline{u}(3) i g_e \gamma^\nu v(4)]$$

$$\times (2\pi)^4 \delta^{(4)}(p_1 + p_2 - q)(2\pi)^4 \delta^{(4)}(q - p_3 - p_4)$$

[37] The order is always adjoint $\times$ interaction $\times$ spinor with the momenta going *backwards* from left to right.

$$\mathscr{M}_2 = -\frac{g_e^2}{(p_1 + p_2)^2}[\overline{u}(3)\gamma^\mu v(4)]\left[\overline{v}(2)\gamma_\mu u(1)\right]$$

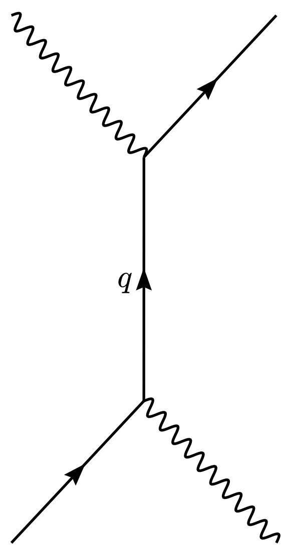
Also, exchanging the incoming e^+ with the outgoing e^- in this diagram we recover the first diagram. Then by rule 8 we find

$$\mathscr{M} = -\frac{g_e^2}{(p_1 - p_3)^2}[\overline{u}(3)\gamma^\mu u(1)]\left[\overline{v}(2)\gamma_\mu v(4)\right]$$

$$+\frac{g_e^2}{(p_1 + p_2)^2}[\overline{u}(3)\gamma^\mu v(4)]\left[\overline{v}(2)\gamma_\mu u(1)\right]$$

7.9. Compton scattering

We again have two diagrams. The first diagram is



We find

$$\mathscr{M}_1 \sim \int \frac{\mathrm{d}^4 q}{(2\pi)^4}\epsilon_\mu(2)\left[\overline{u}(4) i g_e \gamma^\mu \frac{i(\not{q} + mc)}{q^2 - m^2c^2} i g_e \gamma^\nu u(1)\right]\epsilon_\nu(3)^*$$

$$\times (2\pi)^4 \delta^{(4)}(p_1 - p_3 - q)(2\pi)^4 \delta^{(4)}(p_2 + q - p_4)$$

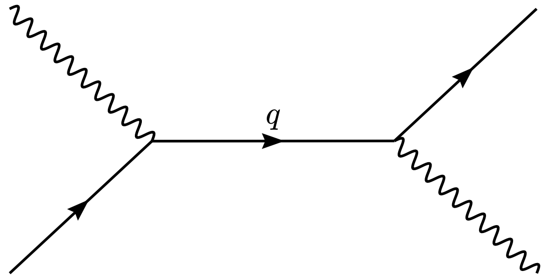
Note, we contract each ϵ_μ with the index of the vertex where it was *created*. Then

$$\mathscr{M}_1 = \frac{g_e^2}{(p_1 - p_3)^2 - m^2c^2}[\overline{u}(4)\not{\epsilon}(2)(\not{p}_1 - \not{p}_3 + mc)\not{\epsilon}(3)^* u(1)]$$

where we have defined

$$\not{a} \equiv \gamma^\mu a_\mu$$

Note, we will write $\not{\epsilon}^* = \gamma^\mu(\epsilon_\mu^*)$. The second diagram is



and gives

$$\mathscr{M}_2 = \frac{g_e^2}{(p_1+p_2)^2 - m^2c^2}[\overline{u}(4)\not{\epsilon}(3)^*(\not{p}_1 + \not{p}_2 + mc)\not{\epsilon}(2)u(1)]$$

Then

$$\mathscr{M} = \mathscr{M}_1 + \mathscr{M}_2$$

7.10. Computing $|\mathscr{M}|^2$

We typically do not know s_i and s_f exactly. We would then need

$$\langle|\mathscr{M}|^2\rangle \equiv \text{average over } s_i \text{ and sum over } s_f \text{ of } \left|\mathscr{M}(s_i \to s_f)\right|^2$$

We would prefer computing $\langle|\mathscr{M}|^2\rangle$ directly. Consider $e^-\mu^-$ scattering. Then

$$|\mathscr{M}|^2 = \frac{g_e^4}{(p_1-p_3)^4}[\overline{u}(3)\gamma^\mu(1)]\left[\overline{u}(4)\gamma_\mu u(2)\right][\overline{u}(3)\gamma^\nu u(1)]^*[\overline{u}(4)\gamma_\nu u(2)]^*$$

We see terms of the form

$$G \equiv [\overline{u}(a)\Gamma_1 u(b)][\overline{u}(a)\Gamma_2 u(b)]^*$$

with a and b representing momenta and spin, while Γ_1 and Γ_2 are 4×4 matrices. We have

$$\begin{aligned}
[\overline{u}(a)\Gamma_2 u(b)]^* &= \left[u(a)^\dagger\gamma^0\Gamma_2 u(b)\right]^\dagger \\
&= u(b)^\dagger\Gamma_2^\dagger\gamma^0 u(a) \\
&= u(b)^\dagger\gamma^0\gamma^0\Gamma_2^\dagger\gamma^0 u(a) \\
&= \overline{u}(b)\overline{\Gamma}_2 u(a)
\end{aligned}$$

with

$$\overline{\Gamma}_2 \equiv \gamma^0 \Gamma_2^\dagger \gamma^0$$

Then

$$G = [\overline{u}(a)\Gamma_1 u(b)]\left[\overline{u}(b)\overline{\Gamma}_2 u(a)\right]$$

We sum over b orientations

$$\sum_{s_b} G = \overline{u}(a)\Gamma_1 \left[\sum_{s_b} u^{(s_b)}(p_b)\overline{u}^{(s_b)}(p_b)\right]\overline{\Gamma}_2 u(a)$$

$$\overset{\text{completeness}}{=} \overline{u}(a)\Gamma_1(\not{p}_b + m_b c)\overline{\Gamma}_2 u(a)$$

$$= \overline{u}(a) Q u(a)$$

with

$$Q \equiv \Gamma_1(\not{p}_b + m_b c)\overline{\Gamma}_2$$

We sum over a orientations

$$\sum_{s_a}\sum_{s_b} G = \sum_{s_a} \overline{u}^{(s_a)}(p_a) Q u^{(s_a)}(p_a)$$

$$= \sum_{s_a}\sum_{i,j=1}^{4} \overline{u}^{(s_a)}(p_a)_i Q_{ij} u^{(s_a)}(p_a)_j$$

$$= \sum_{i,j=1}^{4} Q_{ij}\left[\sum_{s_a} u^{(s_a)}(p_a)\overline{u}^{s_a}(p_a)\right]_{ji}$$

$$= \sum_{i,j=1}^{4} Q_{ij}(\not{p}_a + m_a c)_{ji}$$

$$= \operatorname{tr}[Q(\not{p}_a + m_a c)]$$

implying

$$\sum_{\text{spins}} [\overline{u}(a)\Gamma_1 u(b)][\overline{u}(a)\Gamma_2 u(b)]^* = \operatorname{tr}\left[\Gamma_1(\not{p}_b + m_b c)\overline{\Gamma}_2(\not{p}_a + m_a c)\right]$$

which is sometimes called *Casimir's trick*. When either u is replaced by a v the corresponding m_i switches sign.

Again for $e^-\mu^-$ scattering we have $\Gamma_2 = \gamma^\nu$ so

$$\overline{\Gamma}_2 = \gamma^0 \gamma^{\nu\dagger} \gamma^0 = \gamma^\nu$$

so applying Casimir's trick twice we find

$$\langle |\mathscr{M}|^2 \rangle = \frac{g_e^4}{4(p_1 - p_3)^4} \operatorname{tr}[\gamma^\mu (\not{p}_1 + mc) \gamma^\nu (\not{p}_3 + mc)] \operatorname{tr}\left[\gamma_\mu (\not{p}_2 + Mc) \gamma_\nu (\not{p}_4 + Mc)\right]$$

where $m = m_e$ and $M = m_\mu$. We include $\frac{1}{4}$ since we average over all initial spins, i.e. from 2 particles $\times$ 2 orientations.

Casimir's trick reduces the computation of $|\mathscr{M}|^2$ to computing traces. Note,

$$\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$$

$$\operatorname{tr} \alpha A = \alpha \operatorname{tr} A$$

$$\operatorname{tr} AB = \operatorname{tr} BA$$

Recall,

$$\eta_{\mu\nu} \eta^{\mu\nu} = 4$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \Rightarrow \{\not{a}, \not{b}\} = 2a \cdot b$$

These imply the *contraction theorems*,

$$\gamma_\mu \gamma^\mu = 4$$

$$\gamma_\mu \gamma^\nu \gamma^\mu = -2\gamma^\nu \Rightarrow \gamma_\mu \not{a} \gamma^\mu = -2\not{a}$$

$$\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\mu = 4\eta^{\nu\lambda} \Rightarrow \gamma_\mu \not{a} \not{b} \gamma^\mu = 4a \cdot b$$

$$\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\lambda \gamma^\nu \Rightarrow \gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\not{a}\not{b}\not{c}$$

and the *trace theorems*,

$$\operatorname{tr} \text{odd } \#\gamma^\mu = 0$$

$$\operatorname{tr} 1 = 4$$

$$\operatorname{tr} \gamma^\mu \gamma^\nu = 4\eta^{\mu\nu} \Rightarrow \operatorname{tr} \not{a}\not{b} = 4a \cdot b$$

$$\operatorname{tr} \gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma = 4\left(\eta^{\mu\nu}\eta^{\lambda\sigma} - \eta^{\mu\lambda}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\lambda}\right)$$

$$\Rightarrow \operatorname{tr} \not{a}\not{b}\not{c}\not{d} = 4[(a \cdot b)(c \cdot d) - (a \cdot c)(b \cdot d) + (a \cdot d)(b \cdot c)]$$

With $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ we find

$$\operatorname{tr}\gamma^5 = 0$$

$$\operatorname{tr}\gamma^5\gamma^\mu\gamma^\nu = 0 \Rightarrow \operatorname{tr}\gamma^5 \not{a}\not{b} = 0$$

$$\operatorname{tr}\gamma^5\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma = 4i\epsilon^{\mu\nu\lambda\sigma} \Rightarrow \operatorname{tr}\gamma^5 \not{a}\not{b}\not{c}\not{d} = 4i\epsilon^{\mu\nu\lambda\sigma} a_\mu b_\nu c_\lambda d_\sigma$$

As an example we compute the traces in $e^-\mu^-$ scattering

$$\operatorname{tr}[\gamma^\mu(\not{p}_1 + mc)\gamma^\nu(\not{p}_3 + mc)] = \operatorname{tr}\gamma^\mu\not{p}_1\gamma^\nu\not{p}_3 + mc\underbrace{[\operatorname{tr}\gamma^\mu\not{p}_1\gamma^\nu + \operatorname{tr}\gamma^\mu\gamma^\nu\not{p}_3]}_{0 \text{ by } \#\gamma^\mu \text{ being odd}} + (mc)^2\underbrace{\operatorname{tr}\gamma^\mu\gamma^\nu}_{4\eta^{\mu\nu}}$$

We compute the first term by

$$\begin{aligned}
\operatorname{tr}\gamma^\mu\not{p}_1\gamma^\nu\not{p}_3 &= \overbrace{(p_1)_\lambda (p_3)_\sigma}^{\text{numbers}} \operatorname{tr} \overbrace{\gamma^\mu\gamma^\lambda\gamma^\nu\gamma^\sigma}^{\text{matrices}} \\
&= (p_1)_\lambda (p_3)_\sigma 4\big(\eta^{\mu\lambda}\eta^{\nu\sigma} - \eta^{\mu\nu}\eta^{\lambda\sigma} + \eta^{\mu\sigma}\eta^{\lambda\nu}\big) \\
&= 4[p_1^\mu p_3^\nu - \eta^{\mu\nu}(p_1 \cdot p_3) + p_3^\mu p_1^\nu]
\end{aligned}$$

implying

$$\operatorname{tr}[\gamma^\mu(\not{p}_1 + mc)\gamma^\nu(\not{p}_3 + mc)] = 4\big\{p_1^\mu p_3^\nu + p_3^\mu p_1^\nu + \eta^{\mu\nu}[(mc)^2 - (p_1 \cdot p_3)]\big\}$$

The second trace is similar so

$$\begin{aligned}
\langle|\mathscr{M}|^2\rangle &= \frac{4g_e^4}{(p_1 - p_3)^4}\big\{p_1^\mu p_3^\nu + p_3^\mu p_1^\nu + \eta^{\mu\nu}[(mc)^2 - (p_{1\kappa} p_3^\kappa)]\big\} \times \big\{p_{2\mu}p_{4\nu} + p_{4\mu}p_{2\nu} + \eta_{\mu\nu}[(Mc)^2 - (p_2 \cdot p_4)]\big\} \\
&= \frac{8g_e^4}{(p_1 - p_3)^4}\Big[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(Mc)^2 - (p_2 \cdot p_4)(mc)^2 + 2\big(mMc^2\big)^2\Big]
\end{aligned}$$

7.11. Computing Γ and σ

Consider $e^-\mu^-$ scattering with $M \gg m$. We assume the recoil of μ^- can neglected, and would like to determine the cross-section in the lab frame with M at rest. We know

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\hbar}{8\pi Mc}\right)^2 \langle|\mathscr{M}|^2\rangle$$

We have

$$p_1 = \left(\frac{E}{c}, \boldsymbol{p}_1\right); \quad p_2 = (Mc, \boldsymbol{0}); \quad p_3 = \left(\frac{E}{c}, \boldsymbol{p}_3\right); \quad p_4 = (Mc, \boldsymbol{0})$$

and $|\boldsymbol{p}_1| = |\boldsymbol{p}_3| = |\boldsymbol{p}|$ with $\boldsymbol{p}_1 \cdot \boldsymbol{p}_3 = \boldsymbol{p}^2 \cos\theta$. Then

$$(p_1 - p_3)^2 = -(\boldsymbol{p}_1 - \boldsymbol{p}_3)^2 = -4\boldsymbol{p}^2 \sin^2\frac{\theta}{2}$$

$$(p_1 \cdot p_3) = m^2c^2 + 2\boldsymbol{p}^2 \sin^2\frac{\theta}{2}$$

$$(p_1 \cdot p_2)(p_3 \cdot p_4) = (p_1 \cdot p_4)(p_2 \cdot p_3) = (ME)^2$$

$$(p_2 \cdot p_4) = (Mc)^2$$

implying

$$\langle|\mathscr{M}|^2\rangle = \left(\frac{g_e^2 Mc}{\boldsymbol{p}^2 \sin^2\theta/2}\right)^2 \left[(mc)^2 + \boldsymbol{p}^2 \cos^2\frac{\theta}{2}\right]$$

Then

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\alpha\hbar}{2\boldsymbol{p}^2 \sin^2\theta/2}\right)^2 \left[(mc)^2 + \boldsymbol{p}^2 \cos^2\frac{\theta}{2}\right]$$

which is called the *Mott formula*. Assuming $\boldsymbol{p}^2 \ll (mc)^2$ we recover the *Rutherford formula*

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{e^2}{2mv^2 \sin^2\theta/2}\right)^2$$

which is quite nice.

However, in QED decays do not exist! We may consider processes like $\pi^0 \to \gamma + \gamma$ as decays, but these are also scatterings $q + \overline{q} \to \gamma + \gamma$. We will consider *pair annihilation* of the form $e^+ + e^- \to 2\gamma$. We find two diagrams with

$$\mathscr{M}_1 = \frac{g_e^2}{(p_1 - p_3)^2 - m^2c^2} \overline{v}(2) \not{\epsilon}_4 (\not{p}_1 - \not{p}_3 + mc) \not{\epsilon}_3 u(1)$$

$$\mathscr{M}_2 = \frac{g_e^2}{(p_1 - p_4)^2 - m^2c^2} \overline{v}(2) \not{\epsilon}_3 (\not{p}_1 - \not{p}_4 + mc) \not{\epsilon}_4 u(1)$$

and $\mathscr{M} = \mathscr{M}_1 + \mathscr{M}_2$. We assume $e^\pm$ are $\sim$ at rest initially. Then

$$p_1 = mc(1, \mathbf{0}); \quad p_2 = mc(1, \mathbf{0}); \quad p_3 = mc(1,0,0,1); \quad p_4 = mc(1,0,0,-1)$$

and

$$(p_1 - p_3)^2 - m^2c^2 = (p_1 - p_4)^2 - m^2c^2 = -2(mc)^2$$

Also,

$$\not{p}_1 \not{\epsilon}_3 = -\not{\epsilon}_3 \not{p}_1 + 2 \underbrace{p_1 \cdot \epsilon_3}_{\text{0 in Coulomb gauge}}$$

$$\not{p}_3 \not{\epsilon}_3 = -\not{\epsilon}_3 \not{p}_3 + 2 \underbrace{p_3 \cdot \epsilon_3}_{\text{0 by Lorenz gauge}}$$

implying

$$\begin{aligned}(\not{p}_1 - \not{p}_3 + mc)\not{\epsilon}_3 u(1) &= \not{\epsilon}_3(-\not{p}_1 + \not{p}_3 + mc)u(1) \\ &\overset{\text{by Dirac}}{=} \not{\epsilon}_3 \not{p}_3 u(1)\end{aligned}$$

similarly

$$(\not{p}_1 - \not{p}_4 + mc)\not{\epsilon}_4 u(1) = \not{\epsilon}_4 \not{p}_4 u(1)$$

Then

$$\mathscr{M} = -\frac{g_e^2}{2(mc)^2} \overline{v}(2)[\not{\epsilon}_4 \not{\epsilon}_3 \not{p}_3 + \not{\epsilon}_3 \not{\epsilon}_4 \not{p}_4] u(1)$$

Note,

$$\not{p}_3 = mc(\gamma^0 - \gamma^3); \quad \not{p}_4 = mc(\gamma^0 + \gamma^3)$$

and

$$\not{\epsilon} = -\boldsymbol{\epsilon} \cdot \boldsymbol{\gamma} = -\begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \boldsymbol{\epsilon} \\ -\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon} & 0 \end{pmatrix}$$

so

$$\not{\epsilon}_3 \not{\epsilon}_4 = -\begin{pmatrix} (\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_3)(\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_4) & 0 \\ 0 & (\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_3)(\boldsymbol{\sigma} \cdot \boldsymbol{\epsilon}_4) \end{pmatrix}$$

with

$$(\boldsymbol{\sigma} \cdot \boldsymbol{a})(\boldsymbol{\sigma} \cdot \boldsymbol{b}) = \boldsymbol{a} \cdot \boldsymbol{b} + i\boldsymbol{\sigma} \cdot (\boldsymbol{a} \times \boldsymbol{b})$$

implying

$$(\not\epsilon_4 \not\epsilon_3 + \not\epsilon_3 \not\epsilon_4) = -2\boldsymbol{\epsilon}_3 \cdot \boldsymbol{\epsilon}_4$$

$$(\not\epsilon_4 \not\epsilon_3 - \not\epsilon_3 \not\epsilon_4) = 2i(\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4) \cdot \boldsymbol{\Sigma}$$

with

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}$$

Then

$$\mathscr{M} = \frac{g_e^2}{mc} \overline{v}(2) \left[(\boldsymbol{\epsilon}_3 \cdot \boldsymbol{\epsilon}_4)\gamma^0 + i(\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4) \cdot \boldsymbol{\Sigma}\gamma^3 \right] u(1)$$

We consider the *singlet state* of $e^- e^+$. Then

$$\mathscr{M}_{\text{singlet}} = \frac{1}{\sqrt{2}} \left(\mathscr{M}_{\uparrow\downarrow} - \mathscr{M}_{\downarrow\uparrow} \right)$$

where $\mathscr{M}_{\uparrow\downarrow}$ is found by using $u^{(1)}$ and $v^{(2)}$ given by

$$u(1) = \sqrt{2mc} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}; \quad \overline{v}(2) = \sqrt{2mc}(0, 0, 1, 0)$$

i.e. spin-up for e^- and spin-down for e^+. Then

$$\overline{v}(2)\gamma^0 u(1) = 0; \quad \overline{v}(2)\boldsymbol{\Sigma}\gamma^3 u(1) = -2mc\hat{z}$$

so

$$\mathscr{M}_{\uparrow\downarrow} = -2ig_e^2 (\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4)_z$$

Likewise,

$$\mathscr{M}_{\downarrow\uparrow} = -\mathscr{M}_{\uparrow\downarrow}$$

implying the amplitude for annihilation of a stationary $e^- e^+$ pair into two γ emerging in $\pm\hat{z}$ is

$$\mathscr{M}_{\text{singlet}} = -2\sqrt{2}ig_e^2 (\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4)_z$$

Note, for the *triplet state* we would have

$$\mathscr{M}_{\text{triplet}} = \frac{1}{\sqrt{2}} \left(\mathscr{M}_{\uparrow\downarrow} + \mathscr{M}_{\downarrow\uparrow} \right) = 0$$

implying it is forbidden! We have

$$\boldsymbol{\epsilon}_+ = -\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\\ 0\end{pmatrix}; \quad \boldsymbol{\epsilon}_- = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\\ 0\end{pmatrix}$$

Also, the γ spins must be opposite so we can have

$$\boldsymbol{\epsilon}_3 = \boldsymbol{\epsilon}_+; \quad \boldsymbol{\epsilon}_4 = \boldsymbol{\epsilon}_-$$

so

$$\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4 = i\hat{k}$$

or in the other case

$$\boldsymbol{\epsilon}_3 \times \boldsymbol{\epsilon}_4 = -i\hat{k}$$

We need the antisymmetric combination[38]

$$\mathscr{M}_{\text{singlet}} = \frac{1}{\sqrt{2}}\left(\mathscr{M}_{\uparrow\downarrow} - \mathscr{M}_{\downarrow\uparrow}\right)$$

$$= -4g_e^2$$

since again $\mathscr{M}_{\text{triplet}} = 0$.

Then in the CM frame

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\hbar c}{8\pi(E_1 + E_2)}\right)^2 \frac{|\boldsymbol{p}_f|}{|\boldsymbol{p}_i|}|\mathscr{M}|^2$$

with

$$E_1 = E_2 = mc^2; \quad |\boldsymbol{p}_f| = mc; \quad |\boldsymbol{p}_i| = mv$$

we find

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{cv}\left(\frac{\hbar\alpha}{m}\right)^2 \Rightarrow \sigma = \frac{4\pi}{cv}\left(\frac{\hbar\alpha}{m}\right)^2$$

We can find τ by considering

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{L}\frac{\mathrm{d}N}{\mathrm{d}\Omega}$$

or $N = L\sigma$ with

[38]Where $\uparrow$ and $\downarrow$ refer to γ polarization.

$$L = \rho v$$

The e^- density for a *single atom* is $|\psi(0)|^2$ so

$$\Gamma \equiv N = \rho v \sigma = v\sigma|\psi(0)|^2 = \frac{4\pi}{c}\left(\frac{\hbar\alpha}{m}\right)^2|\psi(0)|^2$$

The ground state has

$$|\psi(0)|^2 = \frac{1}{\pi}\left(\frac{\alpha mc}{2\hbar}\right)^2$$

so

$$\tau = \Gamma^{-1} = \frac{2\hbar}{\alpha^5 mc^2} \sim 10^{-10}\ \text{s}$$